Institute of Electrical Energy Conversion
Prof. Dr.-Ing. Nejila Parspour
Jun.-Prof. Dr.-Ing. Stefan Mönch

Universität Stuttgart

Research Project

Design and characterization of a 1.2 kV hybrid GaN/SiC four-level ANPC converter

Submitted to the University of Stuttgart by
Karthikeyan Vijayakumar

Study Program:	M.Sc. Electrical Engineering
Matriculation Number:	3696437
IEW Number:	IEW 4013
Starting Date:	15.11.2024
Submission Date:	13.06.2025
Supervisors:	M.Sc. Adrian Söllner
	M.Sc. Ines Bennour
Examiner:	Jun.-Prof. Dr.-Ing. Stefan Mönch

Abstract

Multilevel converters are increasingly vital in high efficiency, high voltage applications such as Electric Vehicle (EV) fast charging, traction inverters, renewable energy systems, electrocaloric heat pumps, etc. By enabling finer voltage steps and distributing voltage stress across multiple devices, these converters offer superior output waveform quality (less harmonic distortion) and reduced switching losses compared to conventional two-level (2L) inverters. While recent research demonstrates converters exceeding six levels for medium voltage applications, scalability to higher voltages remains limited due to the restricted voltage blocking capability of semiconductor devices, especially when high efficiency is a critical goal. Applications requiring operation at very high voltage, such as dielectric actuators and electrocaloric pumps, demand multilevel converter topologies that can provide both high efficiency and voltage scalability. Moreover, multilevel inverters can increase control complexity, PCB design complexity, and overall system cost as a result of the additional power switches and gate driver requirements. Taking these factors into account, this work presents the design, simulation, and experimental validation of a 1.2 kV hybrid GaN/SiC four-level (4L) ANPC converter, specifically optimized for high voltage and high efficiency operation. After comparing DC-NPC, ANPC, and π-type topologies, the ANPC topology was selected for its optimal balance of efficiency, component count, and compatibility with low voltage Wide Bandgap (WBG) devices like GaN HEMTs. The ANPC architecture enables stacking of these high performance devices while also providing enhanced control flexibility and high voltage scalability. GaN devices are used in low voltage switching positions to make use of their fast switching capabilities and lower power loss, while SiC devices are employed in positions requiring high voltage blocking capability. A two-level hysteresis current control algorithm is adopted, enabling dynamic frequency modulation, in contrast to conventional fixed frequency PWM control. The gate signal logic for each switch is carefully derived to ensure efficient operation in 4L mode, aiming to minimize the total number of switching transitions. Subsequently, a hardware prototype of the 4L ANPC converter is implemented on a compact, low parasitic, two layer PCB employing techniques such as vertical two layer layout, optimized component placements, and appropriate high voltage isolations. The hysteresis control algorithm is executed on a System on Chip (SoC) integrated into the same hardware board, enabling fully autonomous real time operation. Experimental validation is performed using an Inductor Capacitor (LC) load under low current conditions. The deadtimes between switching transitions were finely tuned, and Zero Voltage Switching (ZVS) was successfully implemented to achieve highly efficient operation. Experimental results demonstrate an efficiency of 98.84% in 4L operation, compared to 98.03% in 2L mode, with a 41.63% reduction in converter losses. Furthermore, a comparison of voltage ripple in the buffer capacitor voltages is performed for different combinations of buffer and output capacitances, providing a basis for selecting appropriate buffer capacitor values.

Contents

List of Tables V

List of Figures VI

Listings VII

1 Introduction 1
- 1.1 Background . . . 1
- 1.2 Problem and Research Questions . . . 2
- 1.3 Approach and Structure of Work . . . 2

2 Theory on NPC Multilevel Converters 4
- 2.1 Neutral Point Clamped (NPC) Multilevel Converters . . . 4
  - 2.1.1 Diode-Clamped NPC (DC-NPC) . . . 4
  - 2.1.2 Active Neutral Point Clamped (ANPC) . . . 5
  - 2.1.3 π-Type NPC (π-NPC) . . . 6
  - 2.1.4 Why 4L ANPC? . . . 6

3 Simulation of Four Level ANPC converter 10
- 3.1 Goal of the Simulation . . . 10
- 3.2 Hysterisis Two Level Control . . . 11
- 3.3 Four Level Control of ANPC . . . 11
  - 3.3.1 Switch Groups Logic . . . 13
- 3.4 Simulation Results . . . 13
  - 3.4.1 4L Output Waveforms . . . 13
  - 3.4.2 Voltage across Buffer Capacitors . . . 14

4 Hardware Design 16
- 4.1 Circuit Design . . . 17
  - 4.1.1 Power Circuit . . . 17
  - 4.1.2 Gate Driver Circuit . . . 19
  - 4.1.3 SoC and Current Feedback . . . 21
- 4.2 PCB Design . . . 21
  - 4.2.1 Gate Loop Design and Inductance Minimization . . . 21
  - 4.2.2 Power Loop Design and Inductance Minimization . . . 23
- 4.3 Gate Signal Logic Table Implementation in Hardware . . . 26

5 Experiment and Results 27
- 5.1 Experiment Setup . . . 27
- 5.2 Experiment Results . . . 29
  - 5.2.1 Circuit Parameters . . . 29
  - 5.2.2 Zero Voltage Switching and Optimising Deadtime . . . 29

5.2.3 2L and 4L Output waveforms 33
5.2.4 Efficiency Calculation 33
5.2.5 Buffer capacitor selection 36

6 Summary and Conclusion **40**
6.1 Summary 40
6.2 Conclusion 40

Bibliography **42**

List of Tables

2.1 Switching States for Output Voltage Levels 5
2.2 Switching States for Output Voltage Levels 6
2.3 Switching States for Output Voltage Levels 7
2.4 Voltage Blocking Requirements for ANPC and π-Type Topologies in a 1.2 kV System for 4L Operation 8
2.5 Switch Characteristics 8
2.6 π topology voltage paths 8
2.7 ANPC Topology voltage paths 8

3.1 Logic table for switching 4L-ANPC 13
3.2 Operating Conditions for 4L-ANPC Converter Simulation 14

4.1 Comparison between Hall-effect Sensor and TCPA300 Current Probe 21
4.2 Switch Status Based on V_{out} for 2L operation 26

5.1 Summary of Voltage and Current Measurements 29
5.2 Values of Additional Components Used During Measurement 29
5.3 Optimal ZVS Transition: L_0 OFF $\rightarrow$ H_6 ON 29
5.4 Optimal ZVS Transition: H_6 OFF $\rightarrow$ L_0 ON 30
5.5 Operating Parameters and Deadtime Comparison of 2L and 4L Converters . . . 33
5.6 Measurement Probe Characteristics and Power Losses (2L Mode) 35
5.7 Measurement Probe Characteristics and Power Losses (4L Mode) 35
5.8 Voltage Behavior of DC-Link Capacitors in 4-Level ANPC Converter 37

List of Figures

1.1 Voltage output waveforms of 2L, 3L, 5L and 4L with 0V as baseline converters 1

2.1 4L DC-NPC . . . 5
2.2 4L ANPC . . . 6
2.3 π-type NPC . . . 7

3.1 Hysteresis two level control for Multilevel inverter implemented in PLECS . . 10
3.2 4L-ANPC power circuit in PLECS. . . . 12
3.3 Output voltage levels of 4L ANPC converter. . . . 12
3.4 Output waveforms of 4L ANPC Simulation . . . 14
3.5 Voltage across buffer capacitor in 4L ANPC converter simulation . . . 15

4.1 Top view of the 4L ANPC hardware. . . . 16
4.2 Bottom view of the 4L ANPC hardware. . . . 17
4.3 Power circuit schematic of 4L ANPC . . . 18
4.4 Gate driver schematic of GaN MOSFET . . . 19
4.5 PCB Layout top side . . . 22
4.6 PCB Layout bottom side . . . 22
4.7 Gate loop in the designed PCB . . . 23
4.8 Power loops in 4L ANPC . . . 24
4.9 Top and bottom layer layout with aligned current paths. . . . 24
4.10 Power loops 1 and 2 in PCB . . . 25
4.11 Power loop 3 in PCB . . . 25
4.12 Power loop 4 in PCB . . . 25

5.1 Experiment Setup . . . 27
5.2 DUT connections . . . 28
5.3 Experiment setup schematic . . . 28
5.4 Optimised deadtime . . . 30
5.5 Hard switching due to insufficient dead time . . . 31
5.6 Third quadrant loss overshoot due to high dead time . . . 32
5.7 2L Output waveforms . . . 32
5.8 4L Output waveforms . . . 33
5.9 Voltage across buffer capacitors with $C_{\mathrm{OUT}} = 50\ \mu\mathrm{F}$ . . . 38
5.10 Voltage across buffer capacitors with $C_{\mathrm{OUT}} = 5\ \mu\mathrm{F}$ . . . 39

Listings

4.1 Verilog-A Control Logic 26

1 Introduction

1.1 Background

The rapid evolution of modern power systems, electric mobility, and renewable energy technologies has significantly increased the demand for high voltage and highly efficient power conversion. Applications such as Electric Vehicle (EV) fast charging, high power solar inverters, medium voltage motor drives, and energy storage interfaces now commonly operate in the 800–1500 V_{DC} range. This growing demand presents challenges for conventional two-level (2L) converter architectures, which encounter issues such as excessive voltage stress on devices, increased harmonic distortion, and elevated switching frequency requirements as the input voltage rises. To mitigate these issues, multilevel converters can be employed. Unlike conventional 2L converters, which generate output by switching between only two voltage levels ($+V_{DC}$ and $-V_{DC}$), multilevel converters synthesize the output voltage through multiple discrete steps, thereby approximating a sinusoidal output waveform (Fig 1.1).

Multilevel converters provide numerous benefits over traditional 2L converters, especially in high power and high voltage applications. These advantages include:

- **Improved waveform quality:** Multilevel converters can generate output voltages with multiple discrete levels, which more closely approximate a sinusoidal waveform as the number of levels increases. This significantly reduces the Total Harmonic Distortion (THD) in the output, minimizing the need for bulky and expensive filters [1].
- **Lower Electromagnetic Interference (EMI):** Since the voltage transitions between the levels are smaller in multilevel converters compared to 2L converters, the resulting $\frac{dV}{dt}$ is reduced. This minimizes EMI and reduces stress on insulation systems, particularly in motor windings and transformer-based applications [2].
- **Reduced switching losses:** Multilevel converters can achieve high quality voltage output even at lower switching frequencies, leading to reduced switching losses and heat sink size. This enhances system efficiency and improves thermal performance, which is critical in high power applications [3].

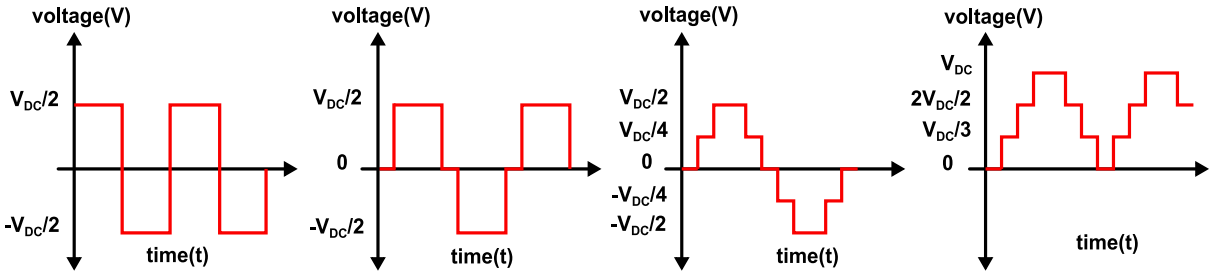


Figure 1.1: Voltage output waveforms of 2L, 3L, 5L and 4L with 0V as baseline converters

- **Use of lower voltage-rated devices:** The total DC bus voltage is divided among multiple semiconductor switches, allowing the use of lower voltage rated devices. This not only reduces cost but also facilitates the integration of Wide Bandgap (WBG) devices such as SiC MOSFETs and GaN HEMTs in high voltage applications [4].
- **Scalability for medium-voltage systems:** Multilevel converter topologies are inherently scalable, making them ideal for high voltage (above 1 kV) and high power applications without requiring step-up transformers or high voltage rated semiconductor switches [5].

Multilevel converters are extensively used in applications that require high power quality, voltage scalability, and efficiency. They are commonly implemented in electric vehicle (EV) drives and fast-charging stations [2]. In industrial environments, MLIs are used in medium-voltage motor drives, compressors, and pumps where modularity and reduced EMI are beneficial [1]. They are also widely adopted in renewable energy systems, such as Photovoltaic (PV) and wind power converters, where compliance with grid codes and high efficiency are critical [4]. Emerging research also highlights their potential in controlling electrocaloric heat pumps for next generation solid state heating and cooling technologies [2], [6].

1.2 Problem and Research Questions

Multilevel converters offer significant advantages over conventional two-level converters. However, their design and implementation present several challenges which must be carefully addressed to realize their full potential.

This research project aims to systematically tackle these challenges by addressing the following key questions:

- Multilevel converters require a large number of semiconductor switches, gate drivers, isolated power supplies and control circuits, which leads to increased system complexity, size, and cost [1]. So, which multilevel topology would be more suitable for medium to high voltage applications to achieve optimal efficiency and reliability?
- The increased number of switching devices complicates control strategies. How can control techniques such as hysteresis current control be effectively applied to multilevel converters to ensure precise and robust operation?
- The PCB design of multilevel converters is inherently complex due to the need for effective voltage isolation between high-voltage power stages and low-voltage control circuits. Additional challenges include minimizing parasitic inductances, integrating multiple isolated feedback signals, and reducing interference between signal and power lines. What design strategies for component placement, signal and power routing, and layout optimization can be employed to achieve an effective PCB design?

1.3 Approach and Structure of Work

The objective of this research is to comprehensively address the above challenges through simulation, design, and experimental validation of an efficient four-level Active Neutral Point

Clamped (4L ANPC) converter. The hysteresis current control algorithm is flashed onto a System on Chip (SoC), integrated on the same PCB, where it processes the current feedback and performs real-time control, providing a compact and complete hardware solution.

- **Chapter 2** presents different types of Neutral Point Clamped multilevel converters with a detailed focus on the Active Neutral Point Clamped (ANPC) converter and its suitability for high-voltage applications.
- **Chapter 3** covers the simulation of the four-level ANPC converter using a two level hysteresis current control strategy. This chapter verifies the gate signal logic table for the 4L ANPC converter, laying the groundwork for future hardware implementation.
- **Chapter 4** details the hardware design process of the converter, including component selection, schematic and PCB design, and the implementation of the gate signal logic table within an existing Verilog program.
- **Chapter 5** presents the experimental results from testing the built hardware in certain operating points. It deals with analysis of output waveforms for both two-level and four-level operation, calculation of efficiency, and compares the performance between the 2L and 4L operation.
- **Chapter 6** presents a summary of the project, highlighting key takeaways and outlining potential improvements and future work aimed at enhancing the design and performance of the 4L ANPC converter system.

2 Theory on NPC Multilevel Converters

The most prominent multilevel topologies include

- Neutral Point Clamped (NPC) converters
- Flying Capacitor (FC) converters
- Cascaded H-Bridge (CHB) converters

Each topology has unique characteristics, advantages, and challenges that make them suitable for different applications [1]. Nowadays, NPC converters are the widely used multilevel converters because of their simple structure, fewer components count, and relatively easy control methodology. So, different types of NPC converter topologies are studied and analysed in this project.

2.1 Neutral Point Clamped (NPC) Multilevel Converters

NPC converters utilize a neutral point or points in the DC bus to generate output voltages that closely approximate a sinusoidal waveform, thereby reducing Total Harmonic Distortion (THD). This is achieved by dividing the DC-link voltage into multiple levels using capacitors and coordinating switching devices with clamping components to connect the output terminal to different voltage points along the DC bus [7]. As a result, multiple voltage levels are generated at the output.

Several variations of the NPC converter topology have been developed, each tailored to suit specific application requirements.

2.1.1 Diode-Clamped NPC (DC-NPC)

The Diode-Clamped NPC converter is one of the earliest multilevel topologies created [8]. In this structure, clamping diodes are employed to limit the voltage stress across switching devices by connecting intermediate DC bus points to specific switching nodes.

An m-level DC-NPC converter, assuming identical voltage blocking capability between the used switches and diodes, requires [7]:

- $2m - 2$ switching devices,
- $m - 1$ input voltage levels (or capacitors),
- $(m - 1)(m - 2)$ clamping diodes.

For example, a four-level DC-NPC converter (Figure 2.1) requires six power switches, six clamping diodes, and three buffer capacitors. These diodes facilitate intermediate voltage steps by connecting the neutral points of the DC-link capacitors to the phase legs, ensuring voltage splitting across switches [1].

Table 2.1 presents the required switch combinations to achieve each output voltage level in a 4L DC-NPC converter. The conduction path depends on the direction of the output current (I_{out}) due to the presence of clamping diodes. The conduction path for the voltage level V_2 is highlighted in red for $I_{\text{out}} > 0$ and in blue for $I_{\text{out}} < 0$ in Figure 2.1. In both cases, current flows through one or more clamping diodes.

The use of clamping diodes in DC-NPC converters leads to increased switching losses and conduction losses, particularly at higher switching frequencies, and the number of diodes required increases significantly with the increase in voltage levels, resulting in reduced efficiency and greater circuit complexity [7]. Consequently, scaling the DC-NPC topology to higher converter levels becomes impractical in many applications.

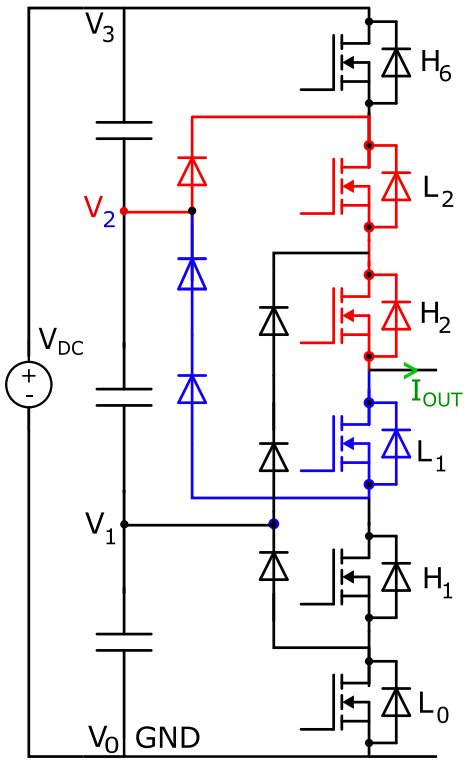


Figure 2.1: 4L DC-NPC

Table 2.1: Switching States for Output Voltage Levels

Voltage level	**Switches ON**
V_3	H_6, L_2, H_2
V_2	L_2, H_2, L_1
V_1	H_2, L_1, H_1
V_0	L_1, H_1, L_0

2.1.2 Active Neutral Point Clamped (ANPC)

To address the limitations of the DC-NPC topology, the Active NPC (ANPC) structure was introduced. In simple ANPC structure configuration, the clamping diodes are replaced with active switching devices such as IGBTs or MOSFETs, while still maintaining the total number of switching devices as in DC-NPC. This allows for flexible switching strategies and improved voltage sharing among the switches [2].

The four-level Hybrid Clamp (4L-HC) topology forms the basis of the ANPC topology. It uses eight switches to realize four-level operation, with each switch handling similar voltage and power levels, enabling the use of identical semiconductor devices [9]. This uniformity enables even loss distribution across the devices, ensuring that no single transistor limits the system's voltage or power capability and allowing full utilization of each device's operating range [10]. However, this configuration can be further optimized by adopting a simplified ANPC structure with six switches, as shown in Figure 2.2. The reduction in the number of switches not only lowers overall system cost and complexity but also contributes to improved efficiency by reducing gate driver losses and control requirements. However, this simplified structure demands that certain switches withstand higher voltage levels.

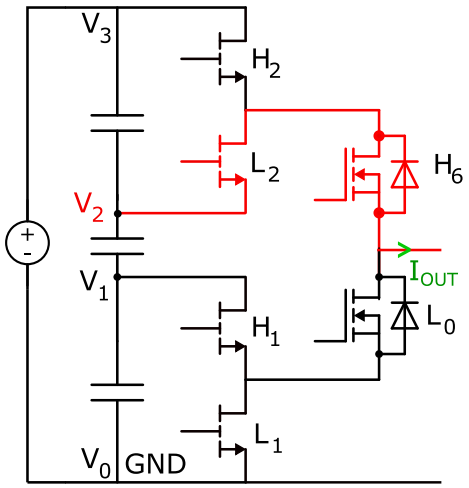


Figure 2.2: 4L ANPC

Table 2.2: Switching States for Output Voltage Levels

Voltage level	**Switches ON**
V_3	H_2, H_6, H_1
V_2	L_2, H_6, H_1
V_1	H_1, L_0, L_2
V_0	L_1, L_0, L_2

Table 2.2 presents the required switch combinations to achieve each output voltage level in a 4L ANPC converter. Unlike DC-NPC, the conduction path does not depend on the direction of the output current (I_{out}) due to the absence of clamping diodes, and only two devices are involved in the path for all voltage levels. The conduction path for the voltage level V_2 is highlighted in red in Figure 2.2.

2.1.3 π-Type NPC (π-NPC)

The π-type NPC converter is a modified variant of ANPC that aims to further improve efficiency and compactness by arranging the switches to obtain shorter current paths [9]. This topology can be simplified and made more efficient further with the usage of Reverse Blocking IGBTs or GaN Monolithic Bidirectional Switches [11].

Table 2.3 presents the required switch combinations to achieve each output voltage level in a 4L π-type NPC converter. The conduction path for the voltage level V_2 is highlighted in red in Figure 2.3.

2.1.4 Why 4L ANPC?

The automotive industry, among others, is scaling DC-link voltages from 400 V to 1.2 kV to reduce losses, increase efficiency, and lighten components. Upgrading to 1.2 kV can reduce

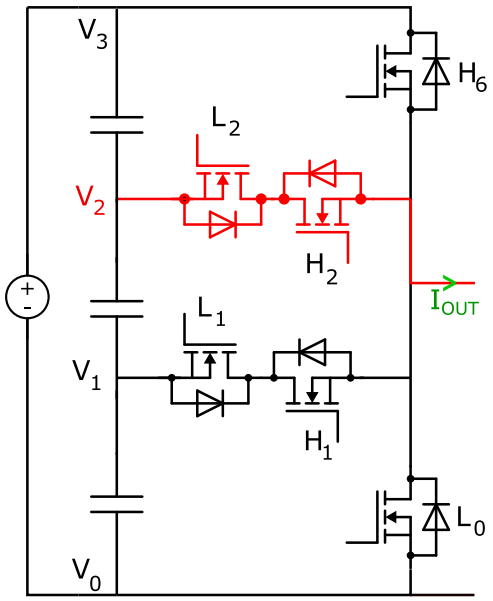


Figure 2.3: π-type NPC

Table 2.3: Switching States for Output Voltage Levels

Voltage level	**Switches ON**
V_3	H_2, H_6, H_1
V_2	L_2, H_2, H_1
V_1	L_2, L_1, H_1
V_0	L_1, L_0, L_2

copper weight by 63% in a 150 kW EV system, which is an additional 30% improvement over 800 V systems [12]. For 350 kW charging, upgrading to voltages above 1 kV also help to keep the cable weight within the safe handling limit of 22.7 kg [12].

Moving towards very high levels in multilevel converter topologies (e.g., five-level or higher) can enhance efficiency and reduce harmonic distortion significantly. However, this often comes at the cost of increased component count, higher costs, reduced power density, increased complexity (hardware design, voltage balancing, modulation, etc.), and decreased reliability [5]. While at least 4L designs are necessary to match the efficiency of lower-voltage two-level systems (e.g., 400 V_{DC} 2L vs. 800 V_{DC} 4L with idealized switch voltage ratings [13]), further increases in level count may offer diminishing returns unless accompanied by novel approaches such as integrated gate drives and efficient thermal designs. Hence, a careful balance must be maintained between performance gains and design complexity, particularly in medium- to high-voltage applications. Therefore, in this project, a four-level topology is chosen as a midway solution, offering enhanced efficiency and reduced harmonic distortion without significantly increasing system complexity or cost.

The ANPC topology offers several advantages over other NPC inverter topologies such as the Diode-Clamped Neutral-Point Clamped (DC-NPC) and the π-type topologies in high-voltage applications.

ANPC vs. DC-NPC

In contrast to the DC-NPC topology, which relies on passive clamping diodes susceptible to reverse recovery losses and localized thermal stress, the ANPC topology employs active switches to dynamically control clamping paths [14]. This active control reduces switching losses, improves thermal distribution across semiconductor devices, and enhances modulation/control flexibility. Furthermore, the ON-state conduction losses of active clamping switches are significantly lower than those of passive diodes, contributing to improved overall efficiency and reliability of the converter.

ANPC vs. π-type NPC

Most switches in the ANPC topology are required to block lower voltages than those in the π-type topology. Consequently, scaling the π-type topology to high voltage is challenging, since most switches (4 out of 6) must withstand voltages near the full DC-link level (Table 2.4).

Table 2.4: Voltage Blocking Requirements for ANPC and π-Type Topologies in a 1.2 kV System for 4L Operation

Topology	Switches	Voltage Blocking Requirement	Technology
ANPC [2]	L_1, L_2, H_1, H_2	33% of DC-link → 400 V + safety margin	650 V GaN
	L_0, H_6	66% of DC-link → 800 V + safety margin	1.2 kV SiC
π-type [9]	L_1, H_2	33% of DC-link (static) , 33% of DC-link + parasitic overshoot (transition) → approx. 400 V + margin	650 V GaN
	L_2, H_1	66% of DC-link (static) , 33% of DC-link + parasitic overshoot (transition) → approx. 800 V + margin	1.2 kV SiC
	L_0, H_6	Full DC-link + safety margin → up to 1.7 kV	1.7 kV SiC

Table 2.5: Switch Characteristics

Switch	V_{DS}(max)	$R_{DS(ON)}$
GS-065-011-1-L [15]	650 V	150 mΩ
IMBG120R116M2H [16]	1.2 kV	115.7 mΩ
IMBF170R450M1 [17]	1.7 kV	450 mΩ

Table 2.6: π topology voltage paths

State	Switches in Path	Total $R_{ds(on)}$
V_0	L_0 (1.7 kV SiC)	450 mΩ
V_1	L_2 (1.2 kV SiC) + H_2 (650 V GaN)	265.7 mΩ
V_2	L_1 (1.2 kV SiC) + H_1 (650 V GaN)	265.7 mΩ
V_3	H_6 (1.7 kV SiC)	450 mΩ

Table 2.7: ANPC Topology voltage paths

State	Switches in Path	Total $R_{ds(on)}$
V_0	H_2 (GaN) + H_6 (1.2 kV SiC)	265.7 mΩ
V_1	L_2 (GaN) + H_6 (1.2 kV SiC)	265.7 mΩ
V_2	H_1 (GaN) + L_0 (1.2 kV SiC)	265.7 mΩ
V_3	L_1 (GaN) + L_0 (1.2 kV SiC)	265.7 mΩ

The difference in switch technologies used to realize each voltage level in the ANPC and π-type topologies results in variations in both conduction and switching losses. An approximate

comparison of conduction losses is made based on the ON-state resistance ($R_{DS(ON)}$) of the switches, as outlined in Tables 2.6 and 2.7. The specific switches employed for this comparison are detailed in Table 2.5.

It is evident from Tables 2.6 and 2.7 that the ANPC topology produces lower conduction losses, due to lower $R_{\mathrm{ds(on)}}$, at certain voltage levels compared to the π topology during 4L operation. The hardware in this project is designed for a 1.2 kV application and hence, the ANPC topology is chosen. This topology offers an optimal balance between performance, reliability, and complexity. Its capability to handle high voltage with lower voltage-rated devices makes it a compelling choice for this design [14].

3 Simulation of Four Level ANPC converter

3.1 Goal of the Simulation

The objective of this simulation of the four-level(4L) ANPC converter is

- To validate the hysteresis two-level current control strategy for 4L operation
- To verify the gate signal logic table of the 4L ANPC inverter, intended for future hardware implementation

The simulation is performed in PLECS.

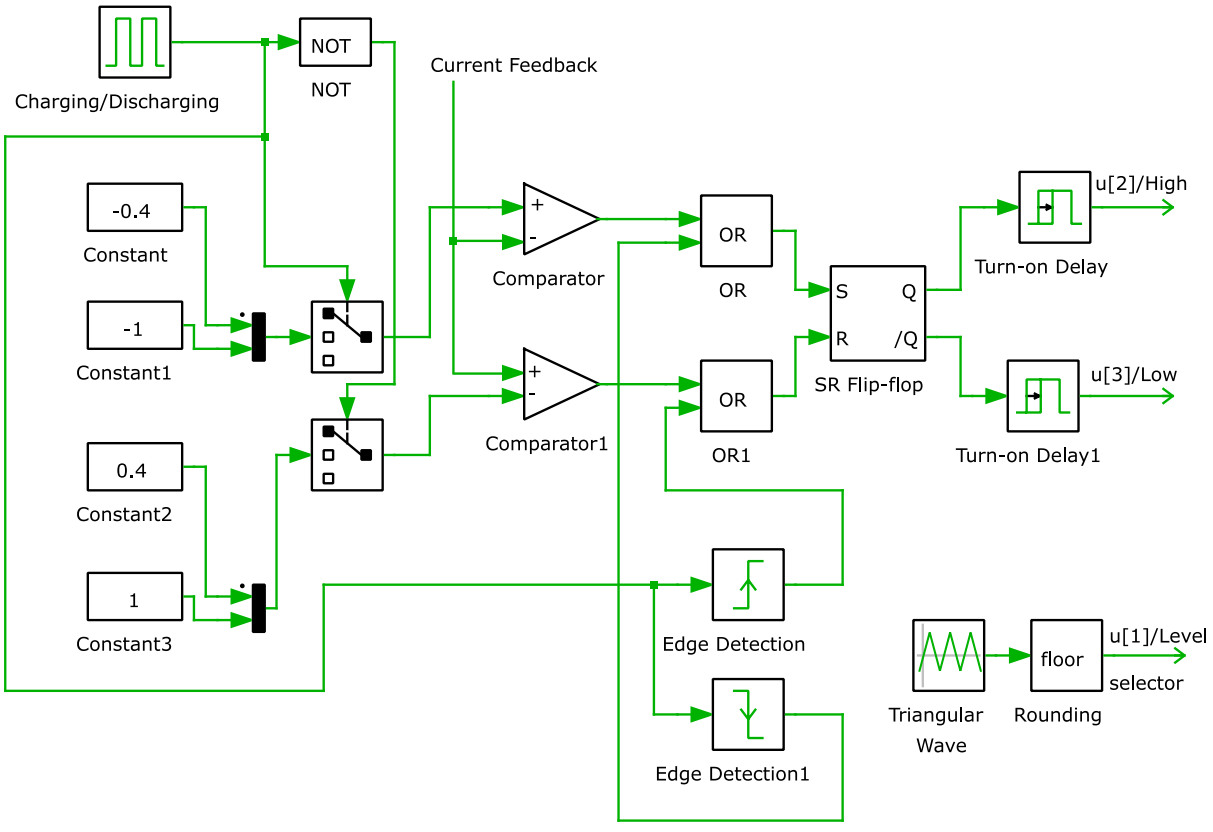


Figure 3.1: Hysteresis two level control for Multilevel inverter implemented in PLECS

3.2 Hysterisis Two Level Control

Analog hysteresis controllers continuously monitor the load current and activate or deactivate specific converter switches when the current reaches predefined upper or lower thresholds. In this way, the load current is maintained within the defined hysteresis band [11]. Unlike traditional pulse-width modulation (PWM) techniques that rely on fixed switching frequencies, hysteresis control operates on an event-driven basis. Switching occurs only when necessary, based on current thresholds, reducing unnecessary switching events and associated losses, especially under varying load conditions.

Hysteresis band limits for load current in this simulation are set differently for charging and discharging cycles of the converter as follows:

- **Charging cycle:** Current limit is set between -0.4 A and 1 A
- **Discharging cycle:** Current limit is set between -1 A and 0.4 A

Comparators H and L perform the hysteresis logic:

- **Comparator H:** Outputs logic 1 if the actual current is less than -0.4 A during charging or less than -1 A during discharging. This generates a high signal (H) at the end of the hysteresis control (Fig 3.1).
- **Comparator L:** Outputs logic 1 if the actual current exceeds 1 A during charging or exceeds 0.4 A during discharging. This generates a low signal(L) at the end of the hysteresis control (Fig 3.1).

Turn-on and turn-off delays are introduced to emulate the dead-time behavior, which is essential for preventing shoot-through conditions and reducing unwanted switching losses (Section 5.2.2). A rounded-off triangular waveform is used as the reference signal for generating the various output levels of the multilevel inverter. These high and low signals at the end of the hysteresis control are sent to the appropriate switches depending on the state and level of operation the inverter currently is in as per Table 3.1.

3.3 Four Level Control of ANPC

The four-level ANPC (4L ANPC) converter consists of six switches (Fig. 3.2). The control signals are provided to these switches according to Table 3.1. In the 4L ANPC converter, the output voltage levels are defined here as:

- $V_0 = 0$ V (zero level)
- $V_1 = \frac{1}{3}V_{\mathrm{DC}}$
- $V_2 = \frac{2}{3}V_{\mathrm{DC}}$
- $V_3 = V_{\mathrm{DC}}$ (full DC-link voltage)

The switching group $\mathbf{S_1}$ handles the transition between V_0 and V_1. Similarly, $\mathbf{S_2}$ and $\mathbf{S_3}$ handle the transitions between V_1 and V_2, and between V_2 and V_3, respectively (see Fig. 3.3).

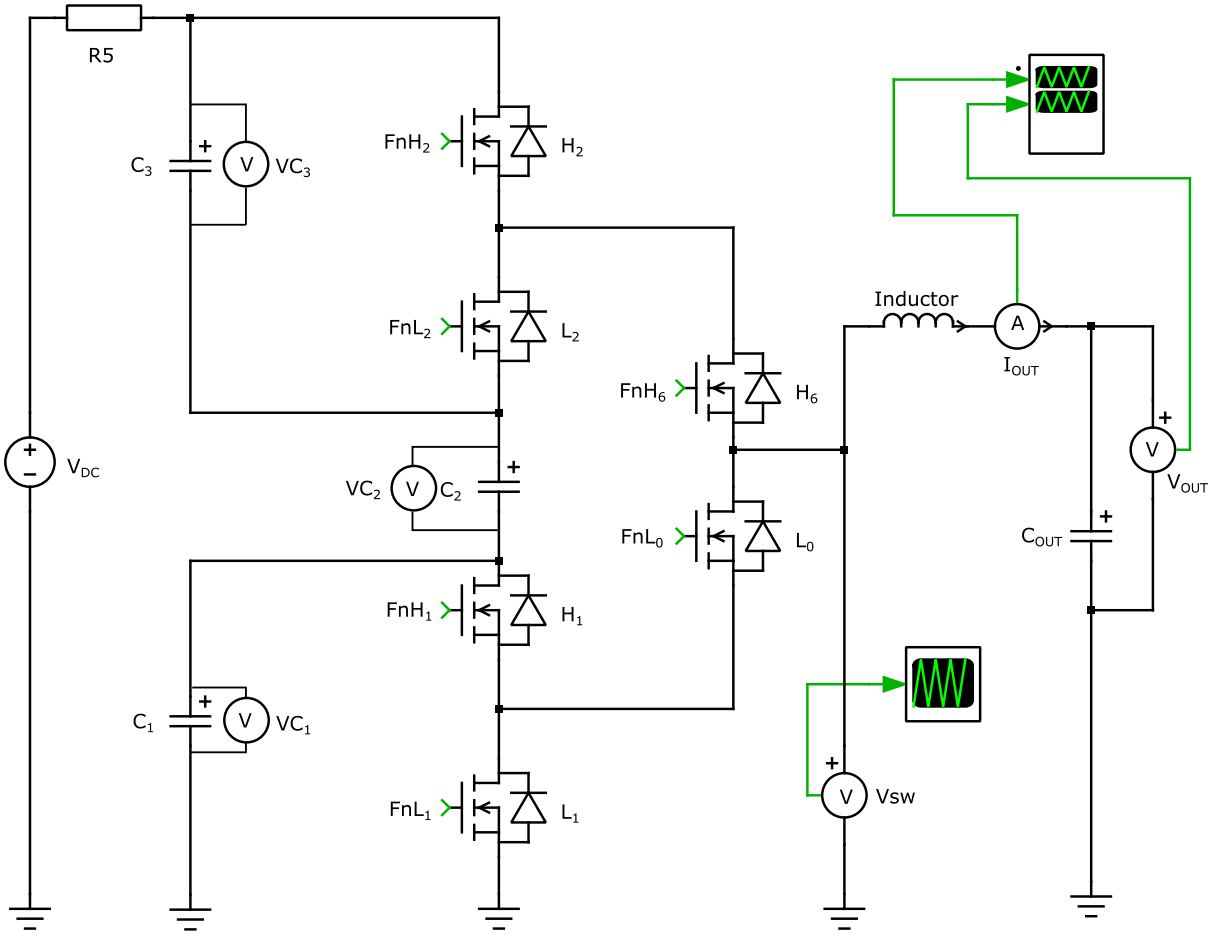


Figure 3.2: 4L-ANPC power circuit in PLECS.

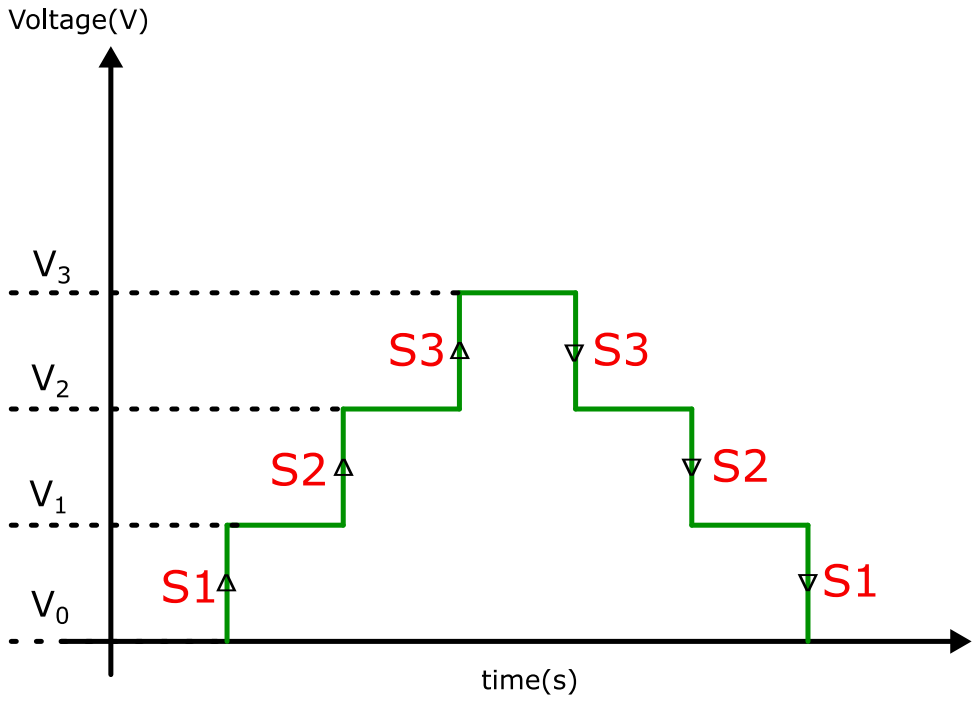


Figure 3.3: Output voltage levels of 4L ANPC converter.

3.3.1 Switch Groups Logic

The main idea behind the logic table is to minimize the number of switching operations during level transitions.

The current path for V_0 level consists of switches L_1 and L_0, while the path for V_1 consists of H_1 and L_0 (Table 2.2). In switching group S_1 (controlling V_0-V_1 transitions), switch L_0 remains continuously ON to minimize switching activity. Switches L_1 and H_1 are toggled based on the required output level (V_0 or V_1). Switch L_2 is also kept ON, as it doesn't affect the control logic, unlike H_6 and H_2, which can hinder performance if activated.

The same principle applies to switching groups S_2 and S_3. The complete switching logic was implemented and verified in PLECS. Table 3.1 specifies the gate signals for each switch. For example, switch L_1 is driven by comparator L output during S_1 operation, while remaining OFF during S_2 and S_3 operations.

This logic is implemented in PLECS as per the "Logic in PLECS" column of the Table 3.1. The selection of the switching group ($\mathbf{S_1}$, $\mathbf{S_2}$, or $\mathbf{S_3}$) is performed by signal **u[1]**, while **u[2]** and **u[3]** correspond to the outputs of comparators **H** (H_i) and **L** (L_i), respectively (Fig. 3.1).

For example, consider the logic expression `(u[1] == 0) && u[3]` for $\mathbf{L_1}$:

- **u[1]==0** selects switching group $\mathbf{S_1}$, and
- **u[3]** (comparator **L** output) is applied to $\mathbf{L_1}$ during this state.

Table 3.1: Logic table for switching 4L-ANPC

Fn	S_1 (N_{1_i})	S_2 (N_{2_i})	S_3 (N_{3_i})	Logic in PLECS
L_1	L	0	0	`(u[1]==0) && u[3]`
H_1	H	1	1	`(u[1]==0) && u[2] || (u[1]==1) || (u[1]==2)`
L_2	1	1	L	`(u[1]==2) && u[3] || (u[1]==1) || (u[1]==0)`
H_2	0	0	H	`(u[1]==2) && u[2]`
L_0	1	L	0	`(u[1]==1) && u[3] || (u[1]==0)`
H_6	0	H	1	`(u[1]==1) && u[2] || (u[1]==2)`

3.4 Simulation Results

The simulation of the 4L-ANPC converter is carried out under the operating conditions mentioned in Table 3.2:

3.4.1 4L Output Waveforms

Figure 3.4 illustrates the output waveforms of the simulated 4L ANPC converter, showing both the output voltage V_{out} and inductor current I_L. The output voltage waveform clearly exhibits multiple discrete levels - approximately 0 V, 33 V, 66 V, and 100 V. This confirms proper four-level operation. The output current waveform demonstrates the I_L oscillating within defined

Table 3.2: Operating Conditions for 4L-ANPC Converter Simulation

Parameter	**Value**
DC-link voltage, V_{DC}	100 V
Hysteresis band (charging)	−0.4 A to 1 A
Hysteresis band (discharging)	−1 A to 0.4 A
System frequency	20 Hz
Load capacitance	20 μF
Load inductance	0.6 mH
Input buffer capacitors	500 μF

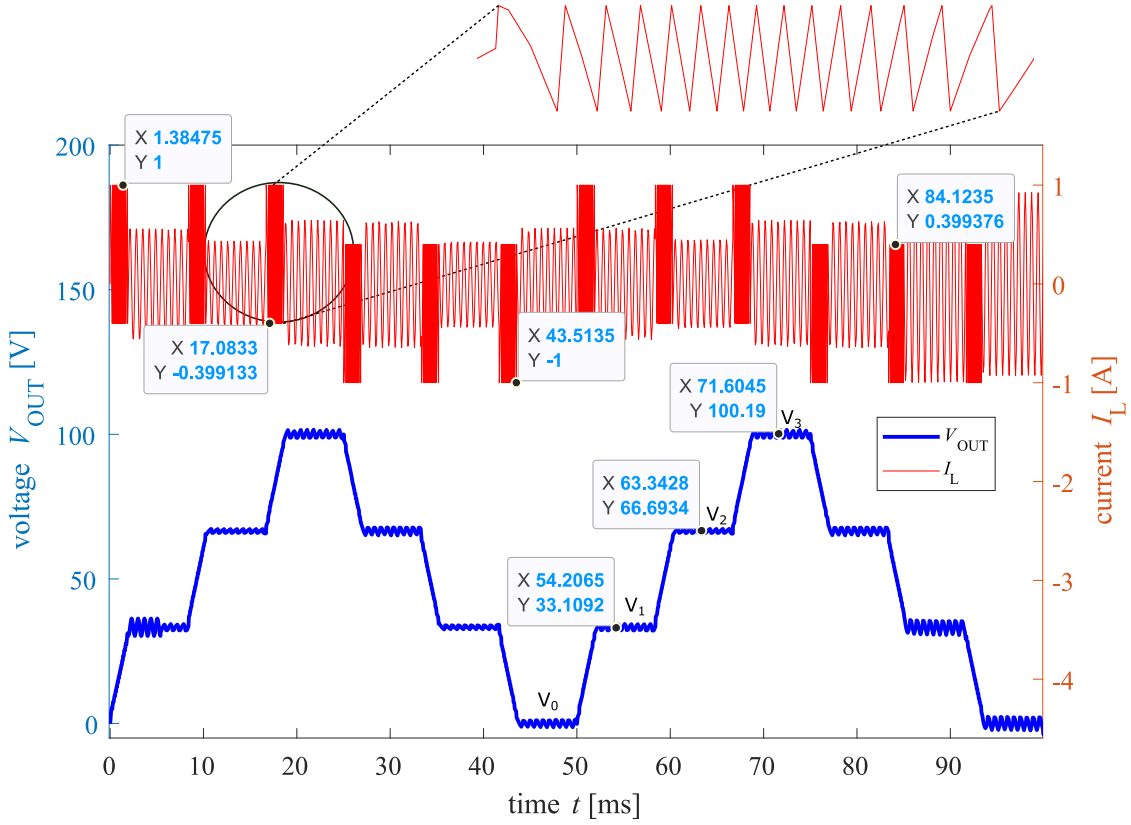


Figure 3.4: Output waveforms of 4L ANPC Simulation

bounds, consistent with hysteresis current control, as described in Table **??**. This control strategy maintains a varying switching frequency (see zoomed view in Figure 3.4), which is a key advantage of using hysteresis control.

3.4.2 Voltage across Buffer Capacitors

Figure 3.5 presents the buffer capacitor voltages (V_{C_1}, V_{C_2}, V_{C_3}) over two complete cycles. In the simulation, the voltages across the buffer capacitors remain relatively balanced throughout the entire period, each maintaining approximately 33.3 V. This behavior is attributed to the significantly lower capacitance of the load capacitor compared to the buffer capacitors. Hence, during the charging and discharging phases, minimal voltage is drawn from the buffer capacitors to the

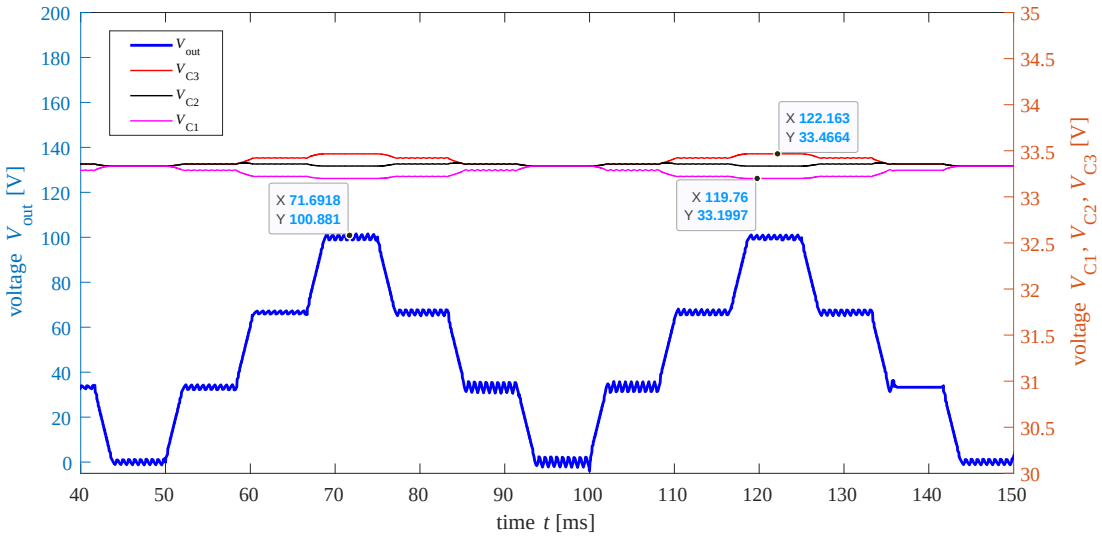


Figure 3.5: Voltage across buffer capacitor in 4L ANPC converter simulation

load. However, in the experimental setup, these voltages exhibit significant fluctuations due to the use of capacitors with comparable capacitance values on both the buffer and load sides.

4 Hardware Design

The 4L ANPC converter hardware is designed and built for 1.2 kV operation. The top and bottom views of the fabricated hardware are shown in Fig. 4.1 and Fig. 4.2, respectively.

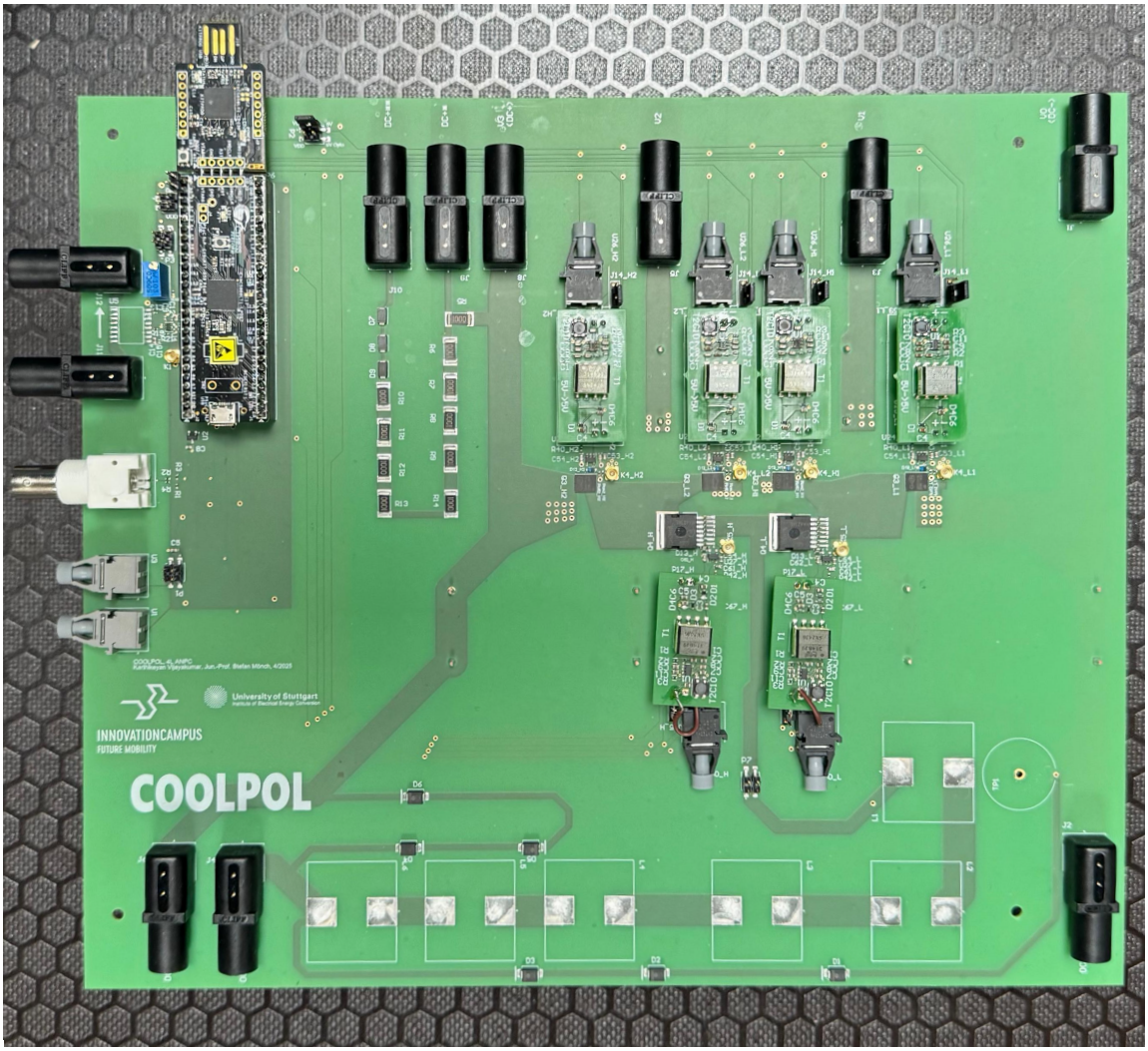


Figure 4.1: Top view of the 4L ANPC hardware.

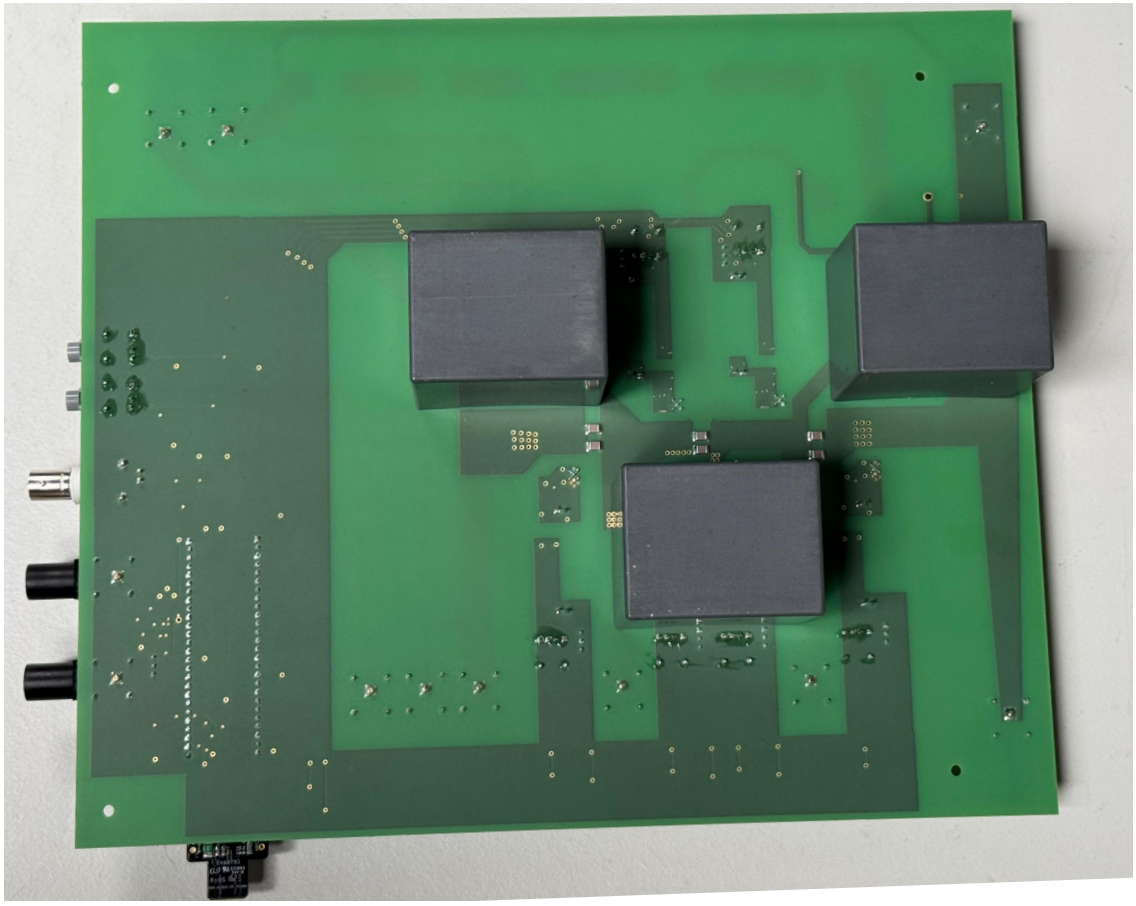


Figure 4.2: Bottom view of the 4L ANPC hardware.

4.1 Circuit Design

The circuit design of the 4L ANPC converter can be categorized into the following key components:

- Power stage of the 4L ANPC converter,
- Gate driver circuits for SiC and GaN switches, and
- System-on-Chip (SoC) control and current feedback circuitry.

4.1.1 Power Circuit

Power Switches

The ANPC converter is designed for a 1200 V_{DC} input. The topology consists of six switches (Figure 4.3). Due to the distribution of the DC-link voltage across the switches, L_1, L_2, H_1, H_2 are required to block only 400 V each, as discussed in Chapter 2. Therefore, the GaN device *GS-065-011-1-L*, which has a drain-to-source voltage rating of $V_{DS} = 650$ V, is selected. Using

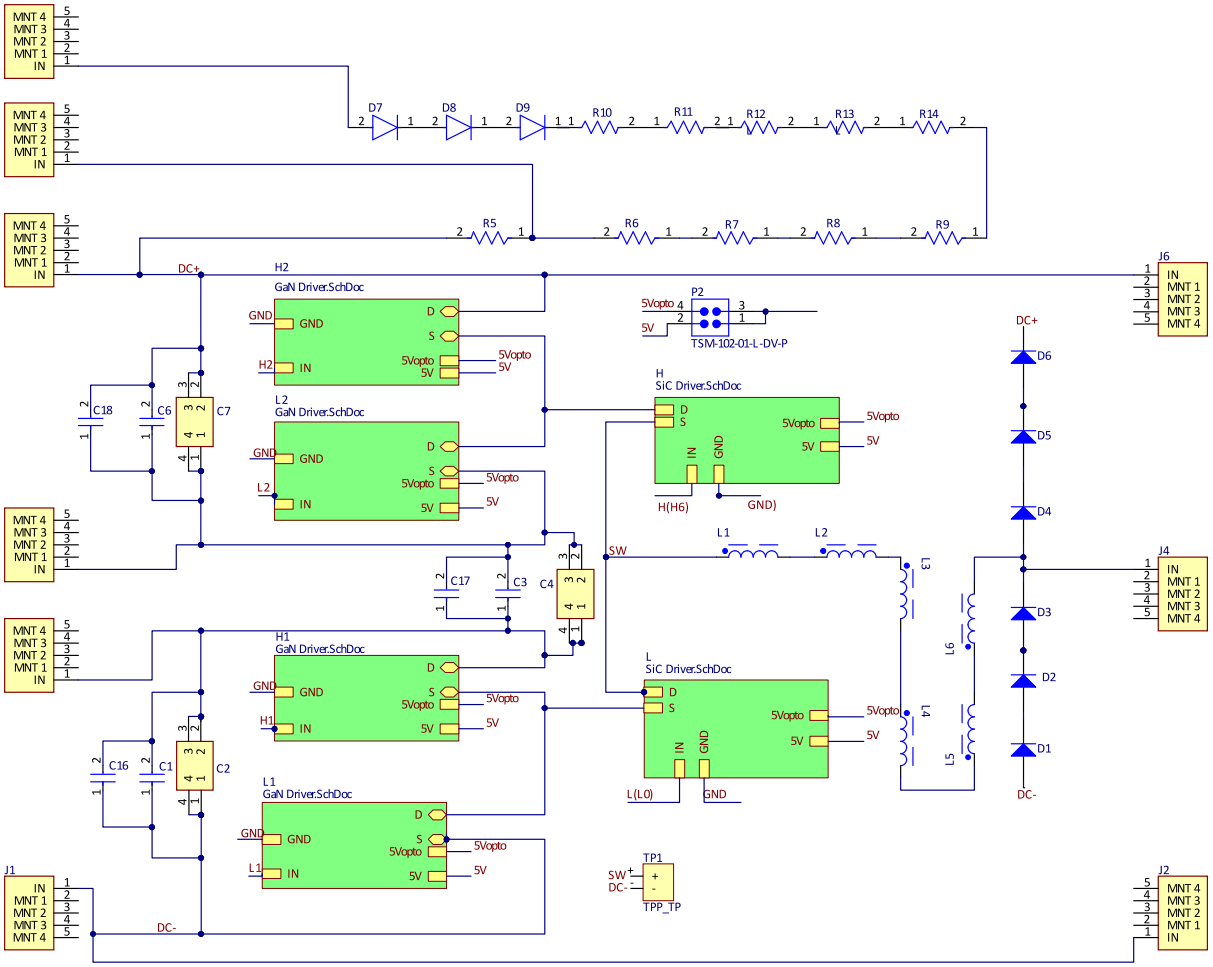


Figure 4.3: Power circuit schematic of 4L ANPC

GaN switches enable higher efficiency and a more compact design due to their inherent advantages, including higher electron mobility, lower on-resistance, faster switching speeds, and reduced parasitic capacitances.

On the other hand, switches L_0 and H_6 must block 800 V each, as also discussed in Chapter 2. Hence, the widely used SiC MOSFET technology is employed. The selected device is *IMBG120R116M2H*, with a voltage rating of $V_{DS} = 1.2\,\text{kV}$, ensuring sufficient blocking capability.

The diodes D_7–D_9 and resistors R_5 to R_{14} are connected at the input from the power source prior to the converter stage. These components prevent reverse current flow and enable accurate measurement of current and voltage in that region (will be discussed in chapter 5).

Capacitances

Three DC-link buffer capacitors are required in the 4L ANPC topology (Figure 4.3). Due to the voltage distribution across the converter levels, each capacitor must be rated above 400 V to ensure reliable operation. The required capacitance value depends on the inverter's power rating, switching frequency, and the type of load under which the inverter operates [18]. For

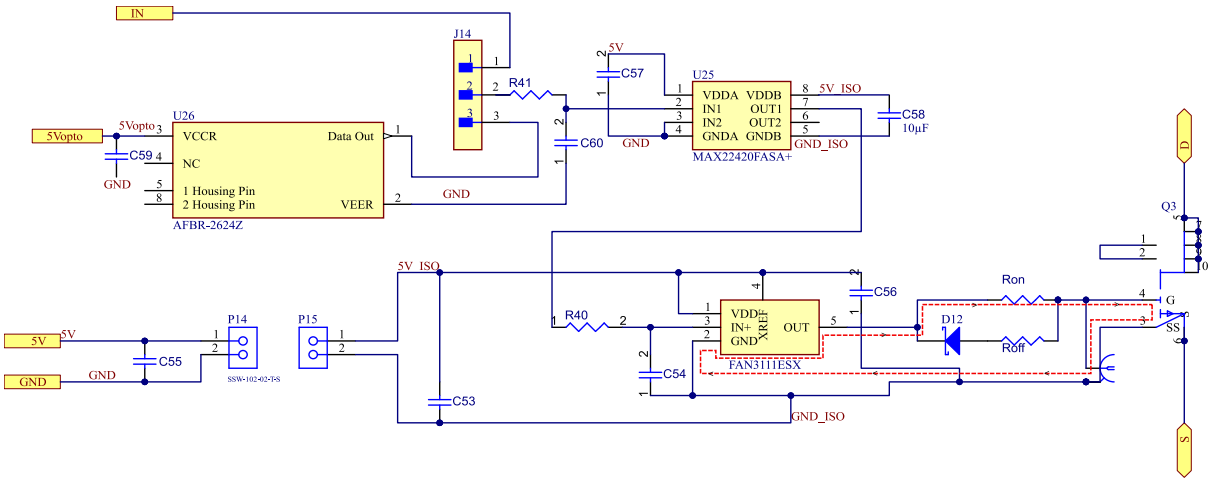


Figure 4.4: Gate driver schematic of GaN MOSFET

this design, the film capacitor *C4AQJEW6110M3AJ* (C_7, C_4, C_2) is selected, offering a 700 V rating and 110 µF capacitance.

Additional ceramic snubber capacitors are connected in parallel with the DC-link capacitors to suppress voltage spikes and high frequency noise. Due to their compact size, these capacitors can be positioned close to the switches on the PCB, thereby reducing power loop inductance.

Load and Output Circuit

The load consists of a series of inductors and a capacitor bank. Diodes D_1–D_6 are incorporated to clamp overvoltages, either above towards the DC+ rail or below towards the DC- rail, thereby protecting the circuit from voltage spikes from the output side. The experimental setup and detailed specifications of the load components are discussed in Chapter 5.

4.1.2 Gate Driver Circuit

Two similar gate driver circuits have been developed, one for GaN HEMTs and the other for SiC MOSFETs. Each switching device is equipped with an independent gate driver to ensure precise and optimized switching behavior.

Input Signal to Gate Driver

The gate driver input can be configured to accept control signals from either an on-board system-on-chip (SoC) or from an external controller via a selectable jumper configuration (J14), enabling two modes of operation forthe converter:

- **Standalone Mode:** Operates as an independent unit with integrated control using the on-board SoC.
- **External Control Mode:** Functions as a dependent unit that receives switching signaks from an external master controller via an optical interface (AFBR-2624Z).

The input control signal is first sent to the digital isolator (MAX22420FASA+), which provides galvanic isolation while offering a high Common-Mode Transient Immunity (CMTI) [19]. It also translates logic levels from 3.3 V to 5 V. The isolated signal is then fed to the gate driver (FAN3111ESX), which delivers a peak current up to 4 A to rapidly drive the MOSFET gate, enabling fast switching and improved efficiency [20].

Filter and Decoupling capacitors

In the gate driver circuit, two RC low-pass filters are implemented for high frequency noise suppression: one at the input stage (after the optical receiver or SoC) using R41 and C60 and another at the gate driver input using R40 and C54.

The cutoff frequency of the RC low-pass filters, with $R = 1$ kΩ (standard practice) and $C = 10$ pF, is calculated using

$$f_c = \frac{1}{2\pi RC}, \tag{4.1}$$

resulting in a cutoff frequency of approximately 15.9 MHz.

This cutoff frequency is significantly higher than the switching frequency used in the system, which is in the range of 10 kHz. Therefore, the filter allows valid control signals to propagate with minimal attenuation while effectively suppressing high-frequency noise and glitches.

Decoupling capacitors of 10 μF, such as C_{59}, C_{55}, and C_{53}, are utilized to mitigate voltage ripple induced by switching currents by acting as local energy sources that stabilize the supply voltage.

R_{on} and R_{off} resistors

R_{on} and R_{off} are gate resistors used to manage the switching characteristics of the power switches during turn-on and turn-off, respectively. A Schottky diode (BAS40) is placed such that during turn-off, it bypasses R_{on} and forces the gate discharge current through R_{off}.

GaN FETs are highly sensitive to gate ringing, so careful selection of R_{on} is required. If R_{on} is too low, it can cause gate overshoot and EMI and if it is too high, it slows down switching and increases losses [21]. Although the recommended R_{on} value is approximately 15 Ω [15], a more conservative avlue of 22.2 Ω is chosen, about 1.5 times the recommendation, to reduce ringing and improve EMI performance. Conversely, R_{off} is kept small (2.2 Ω) to enable fast gate discharge during turn-off. Faster turn-off transitions are generally acceptable for GaN FETs because they reduce the risk of shoot-through and ensure clean switching, provided layout parasitics are well-managed [22].

On the same account, R_{on} and R_{off} of the SiC drivers are selected to be 15 Ω and 2.2 Ω, respectively.

4.1.3 SoC and Current Feedback

The Cypress PSoC 5LP System-on-Chip (SoC) is integrated directly onto the board, thereby eliminating the need for an external controller to generate control signals. This integration enhances system compactness and simplifies the control architecture. The SoC based control circuit is implemented according to the method described in [11]. To acquire feedback for the load current (I_L), the system can utilize one of the two sensing methods: a Hall-effect sensor (MCR1101-5-5) or a Micsig CP1003B current probe.

The Hall sensor is mounted on the PCB and provides a galvanically isolated analog voltage output, making it suitable for embedded, real-time control applications [11]. In contrast, the Micsig CP1003B current probe is externally connected to SoC reading current from the load side. This probe is designed for high-bandwidth and high-accuracy current measurement, making it ideal for experimental validation and transient analysis.

Table 4.1 compares the key specifications of both current measurement techniques.

Table 4.1: Comparison between Hall-effect Sensor and TCPA300 Current Probe

Parameter	**MCR1101-5-5 [23]**	**Micsig CP1003B [24]**
Measurement Range	±5 A	up to ± 30 A
Bandwidth	~1.5 MHz	Up to 100 MHz
Accuracy	±1%RD typical	±1%RD typical
Response Time	300ns	3.5ns
Use Case	Real-time control	Transient/waveform analysis

It is clear from the table that current probe feedback method is preferred for lab-based measurements due to its higher bandwidth and faster response time.

4.2 PCB Design

The 4L-ANPC inverter circuit, together with its control circuitry, is implemented on a compact two-layer PCB. The design operates as an independent, self-contained unit, requiring no external controller for gate drive or logic management. The PCB layout is carefully optimized to minimize gate and power loop inductance, ensure sufficient voltage isolation between components and tracks, maintain appropriate track widths, and reduce signal interference by minimizing the overlap and proximity of power and control signal traces.

4.2.1 Gate Loop Design and Inductance Minimization

The gate driver is designed in such a way that the power switches can be placed in a straight line, as required by the topology of the 4L-ANPC inverter. The functional components of the gate driver are described in detail in section 4.1.2.

The key aspect of the gate driver PCB layout is the minimization of gate loop inductance (red loop in Fig 4.4). The gate loop parasitic inductance can result in gate overvoltage during the

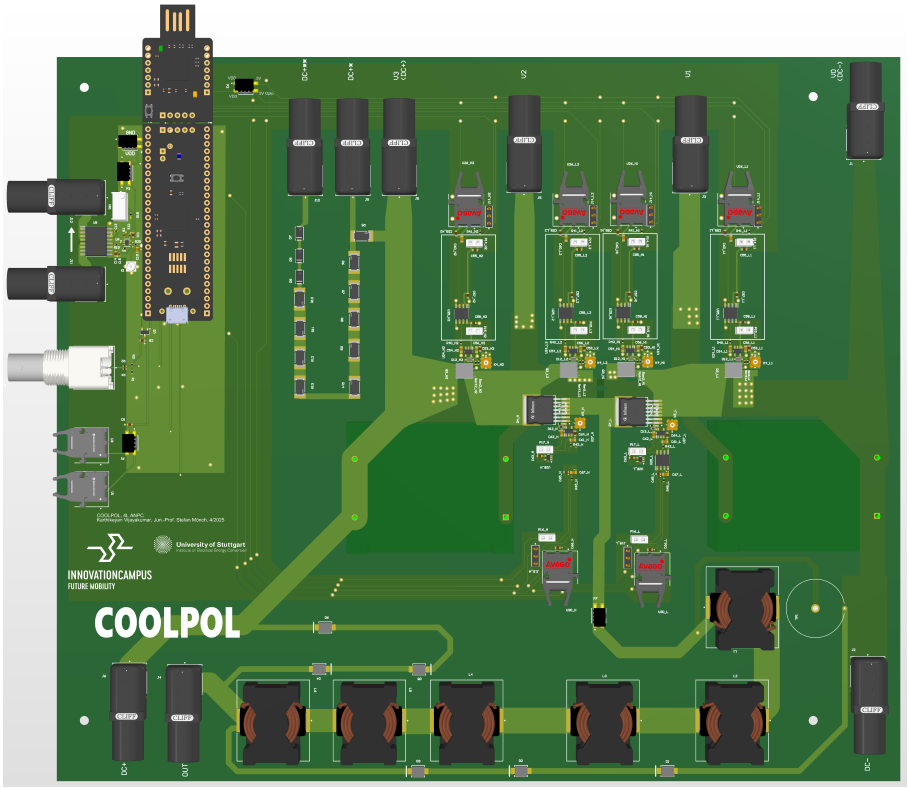


Figure 4.5: PCB Layout top side

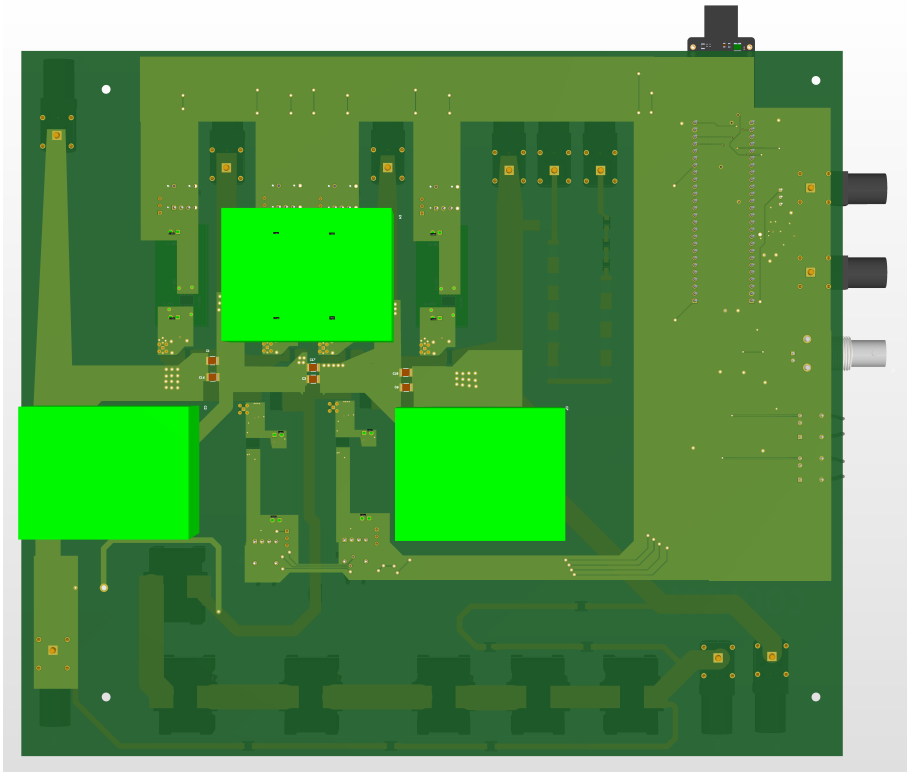


Figure 4.6: PCB Layout bottom side

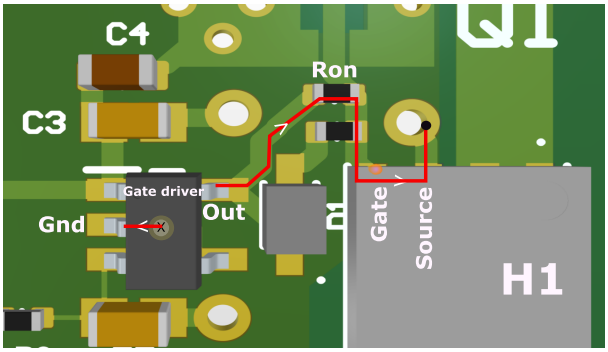


Figure 4.7: Gate loop in the designed PCB

turn ON scenario and unintentional turn ON (false turn ON) during the turn OFF scenario. The following design strategies are implemented to reduce gate loop inductance [22] (Fig 4.7):

- The gate driver ICs and gate resistors are placed as close as possible to the corresponding power switches to minimize loop length.
- Vias are used to route the gate return path to the second PCB layer, which is implemented as a solid copper ground plane. This continuous plane offers a low-impedance path for high-frequency currents, as current naturally follows the path of least impedance [25].
- Decoupling capacitors are placed very close to the power supply pins of the driver ICs to ensure stable voltage levels and suppress high-frequency noise [25].

4.2.2 Power Loop Design and Inductance Minimization

In high-frequency switching applications, such as GaN-based converters, parasitic inductance in the power loop can resonate with the transistor's output capacitance, leading to drain-source voltage ringing [26]. This ringing introduces additional power loss during switching transitions. Parasitic inductance of the power loop is mainly contributed by the PCB tracks, which solely depends on the layout method in the design stage. In the 4-level ANPC topology, multiple power loops exist due to its switch structure (Fig 4.8).

The two common strategies employed for minimizing power loop inductance are [27]:

1. **Loop length minimization**: By reducing the physical distance between components within the current path (loop area) and hence minimizing the inductance.
2. **Magnetic flux cancellation**: By routing current paths in opposing directions on adjacent layers, magnetic fields generated by the currents partially cancel, further reducing effective loop inductance.

Different layout methods are discussed in [27] for the simple totem pole bridge employing these two techniques. In this project, a *vertical layout* is employed for the major power loops, where the input ceramic snubber capacitors are placed on the bottom layer directly beneath the power switches on the top layer. This layout style reduces the overall loop area as the board thickness is significantly smaller compared to lateral trace length in traditional horizontal layouts. The return path is routed directly beneath the forward path (Fig 4.9). Additionally, the

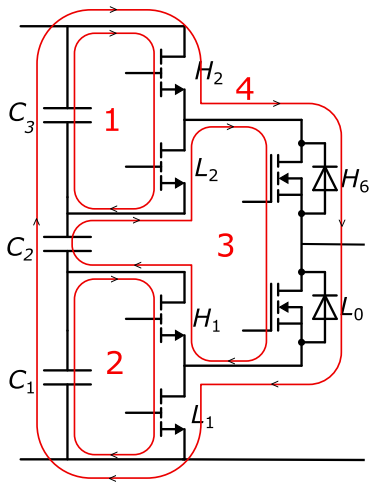


Figure 4.8: Power loops in 4L ANPC

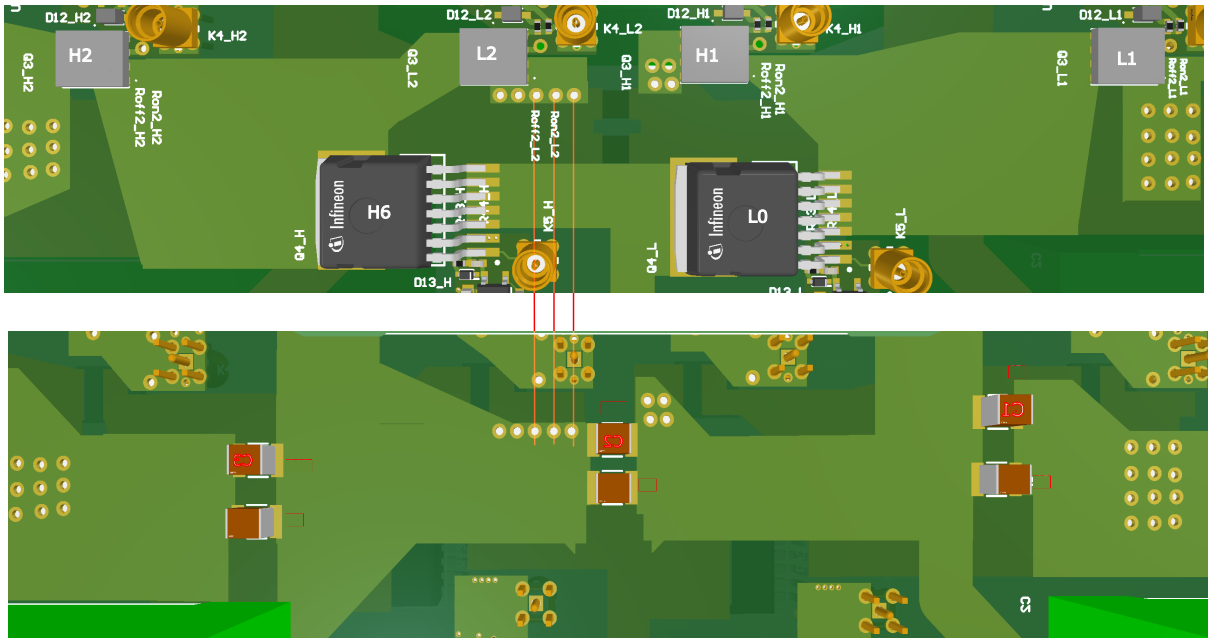


Figure 4.9: Top and bottom layer layout with aligned current paths.

current flowing in opposite directions across the top and bottom layers leads to partial field cancellation, further lowering loop inductance.

These power loops are highlighted in Figures 4.10, 4.11, 4.12, where red lines represent traces on the top layer, blue lines represent those on the bottom layer, and arrows indicate "via" transitions between layers. Loops 1, 2, and 4 are implemented using a vertical layout, ensuring minimal loop area and inductance. In contrast, Loop 3 includes some unavoidable lateral elements due to its specific current path and component placement constraints, resulting in a partial lateral loop structure.

Other improved layout strategies, such as fully vertical configurations with all components on a single side [26], or minimal dual-side layouts with both switches and capacitors placed on the bottom layer [27], introduce significant design complexities in the context of the 4L ANPC topology. These approaches often result in one or more power loops having substantially larger loop areas, thereby increasing parasitic inductance and compromising performance.

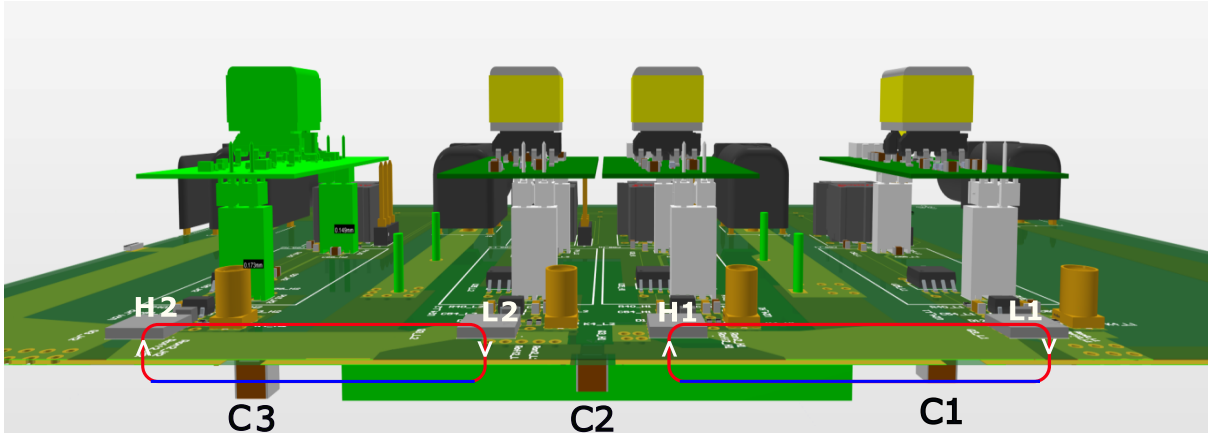


Figure 4.10: Power loops 1 and 2 in PCB

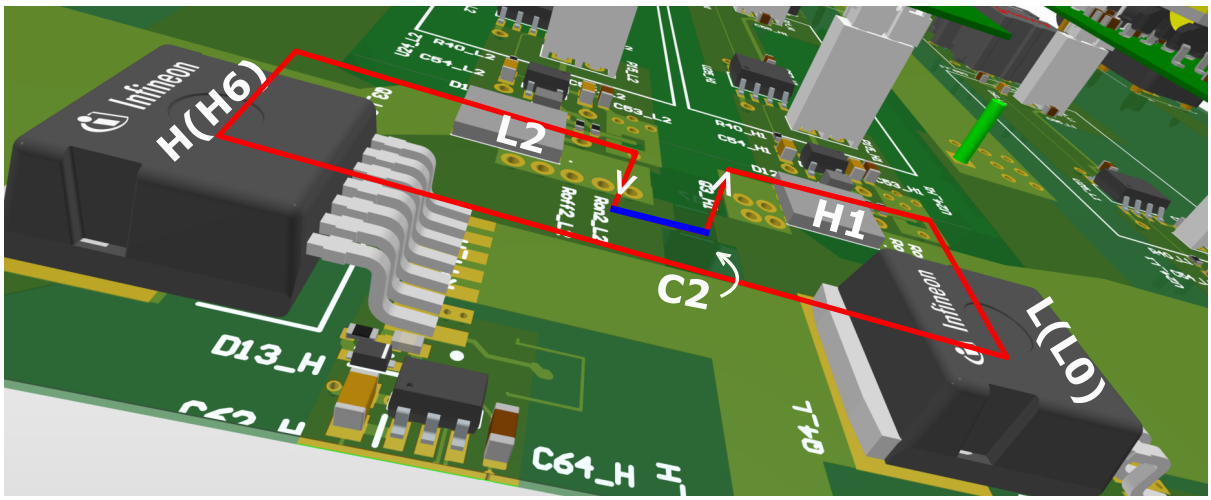


Figure 4.11: Power loop 3 in PCB

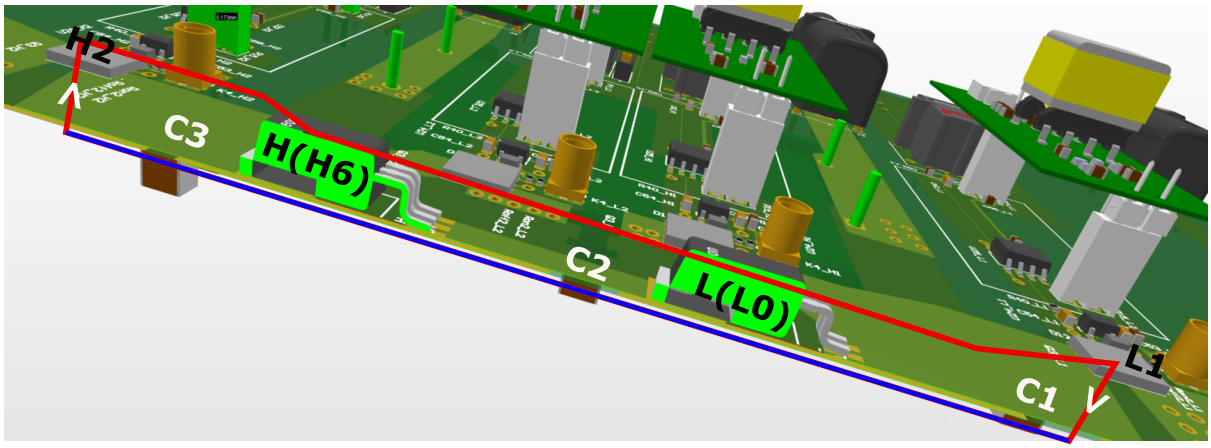


Figure 4.12: Power loop 4 in PCB

4.3 Gate Signal Logic Table Implementation in Hardware

The hysteresis two level control scheme is developed and implemented in hardware as described in [28]. The integrated PSoC is utilized to acquire load current feedback and generate the corresponding switching signals for the gate drivers. The switching logic for four-level (4L) operation is derived from Table 3.1 and implemented in Verilog-A on the PSoC, as shown in Listing 4.1.

The signal `NEN` serves as a mode selector between 2L and 4L operation. When `NEN` is "True", the system operates in 4L mode; otherwise, it defaults to 2L mode. The switching logic for the 2L operation is defined based on the switching states summarized in Table 4.2.

Table 4.2: Switch Status Based on V_{out} for 2L operation

V_{out}	H_2	L_1	H_6	L_0
High	ON	ON	ON	OFF
Low	ON	ON	OFF	ON

```
assign L0 = NEN ? (N2_i & !N3_i & !N1_i & L_i) | (N1_i & !N2_i & !N3_i )
                : L_i;
assign L1 = NEN ? (N1_i & !N2_i & !N3_i & L_i)
                                          : 1;
assign H1 = NEN ? (N1_i & !N2_i & !N3_i & H_i) | (N2_i & !N1_i & !N3_i ) | (
    N3_i & !N1_i & !N2_i ): 0;
assign L2 = NEN ? (N1_i & !N2_i & !N3_i )| (N3_i & !N1_i & !N2_i & L_i) | (
    N2_i & !N1_i & !N3_i ) : 0;
assign H2 = NEN ? (N3_i & !N2_i & !N1_i & H_i)
                                          : 1;
assign L3 = NEN ? 0 : 0;
assign H3 = NEN ? 0 : 0;
assign L4 = NEN ? 0 : 0;
assign H4 = NEN ? 0 : 0;
assign L5 = NEN ? 0 : 0;
assign H5 = NEN ? 0 : 0;
assign H6 = NEN ? (N2_i & !N3_i & !N1_i & H_i) | (N3_i & !N2_i & !N1_i )
                : H_i;
```

Listing 4.1: Verilog-A Control Logic

5 Experiment and Results

The objective of this experiment is to validate the operation of the constructed ANPC 4L hardware (device under test) by capturing and analyzing the input and output waveforms. Subsequently, the system efficiencies in both 2L and 4L operating modes are calculated and compared.

5.1 Experiment Setup

Figure 5.1 shows the complete experimental setup, comprising all major components such as the device under test (DUT), load capacitor, power supply, as well as measurement and display instruments including the oscilloscope and power analyzer.

Figure 5.2 presents a closer look into the DUT connections, including interfaces to the power supply, power analyzer, load capacitor, voltage and current measurement, current feedback, etc.

Figure 5.3 presents a schematic representation of the complete experimental setup, detailing the current and voltage measurement points. The gate drivers are controlled by the SoC (highlighted in pink in Figure 5.3), which generates the appropriate gate signals based on the selected mode of operation (2L or 4L), as per Listing 4.1. A variety of current and voltage probes are used to perform the electrical measurements. The measured quantities and the corresponding measurement probes are listed in Table 5.1.

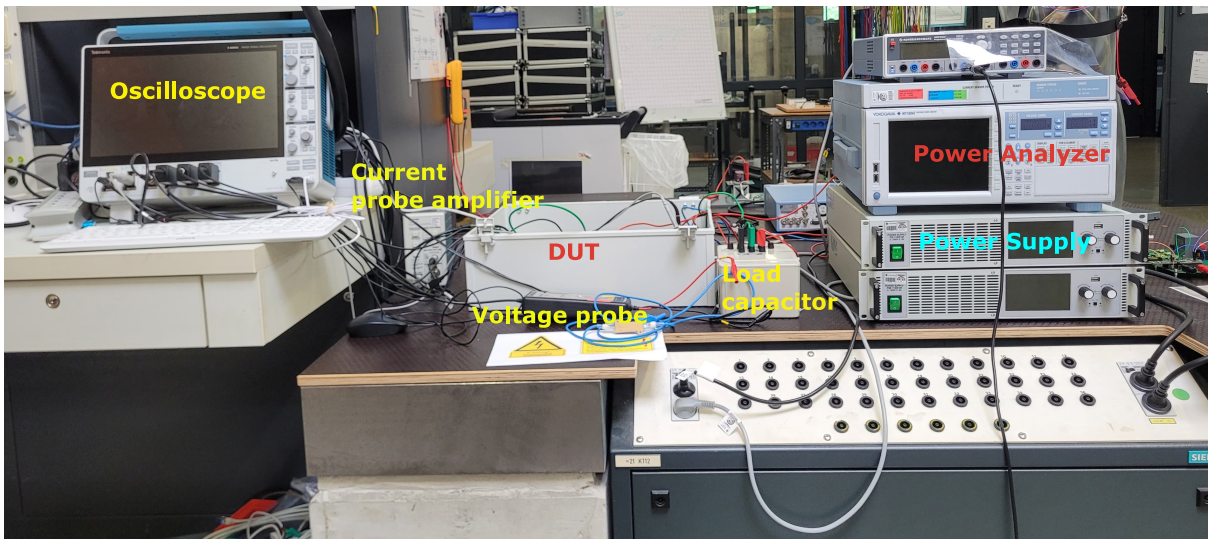


Figure 5.1: Experiment Setup

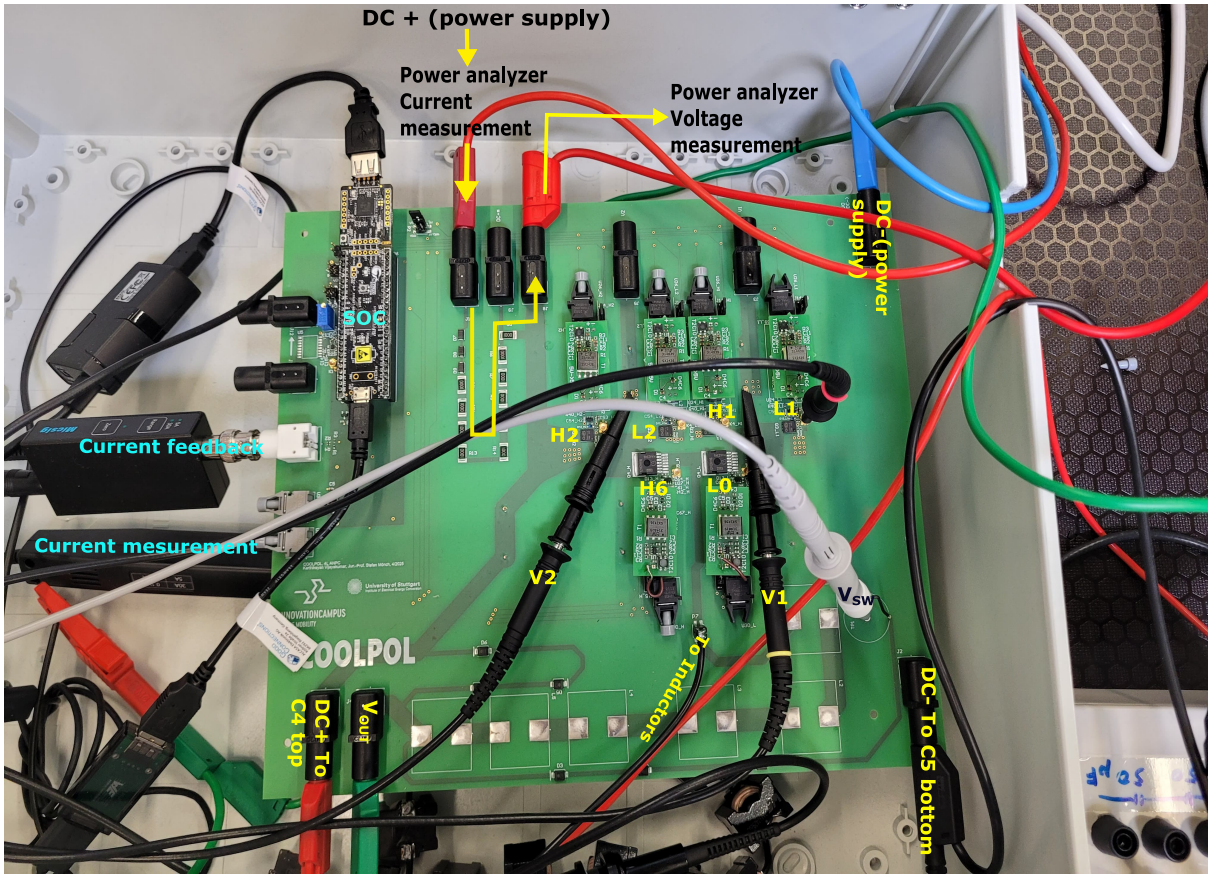


Figure 5.2: DUT connections

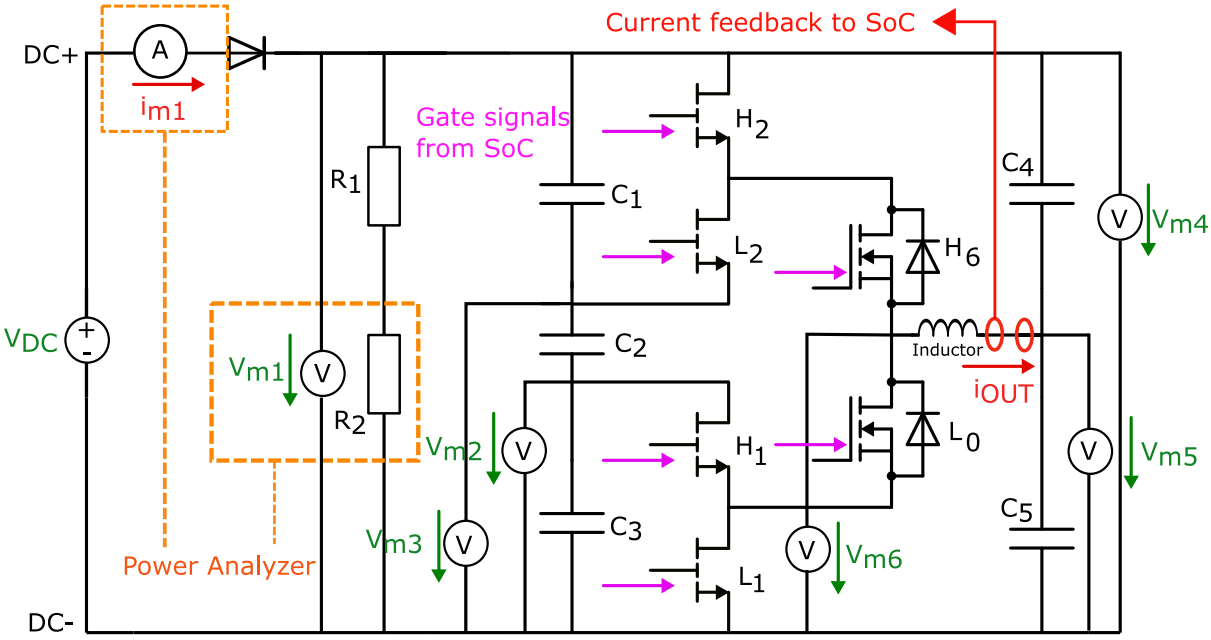


Figure 5.3: Experiment setup schematic

Table 5.1: Summary of Voltage and Current Measurements

Label	Quantity Measured	Measurement Equipment
V_{m1}	DC Input Voltage (V_{DC})	Yokogawa WT1804E Power Analyzer
V_{m2}	Intermediate Node 1 Voltage (V_1)	Tektronix TPP1000 Voltage Probe
V_{m3}	Intermediate Node 2 Voltage (V_2)	Tektronix TPP1000 Voltage Probe
V_{m4}	DC Bus Voltage (V_{DC})	Testec Differential Probe
V_{m5}	Output Voltage (V_{OUT})	Tektronix TPP0850 Voltage Probe
I_{m1}	Input Current (I_{input})	Yokogawa WT1804E Power Analyzer
I_{m2}	Output Current (I_{out})	TCPA300 with TCP305 Current Probe

5.2 Experiment Results

5.2.1 Circuit Parameters

The values of the components within the DUT, such as switches, resistors, and capacitors, are provided in detail in Chapter 4. Table 5.2 lists the values of additional components used during the experiment (Figure 5.2).

Table 5.2: Values of Additional Components Used During Measurement

Component	Description	Value / Rating
R_1	High-ohmic resistor	20 MΩ
R_2	Power analyzer input resistance	2 MΩ
C_1, C_2, C_3	Buffer capacitors	110 µF each
Inductor	Load inductor	14 × 150 µH Litz-wire inductor
C_4, C_5	Load capacitors	50 µF each

5.2.2 Zero Voltage Switching and Optimising Deadtime

The optimum dead times for the rising and falling edges of V_{SW} were experimentally determined by tuning the dead time intervals while monitoring the switch-node voltage (V_{SW}) and inductor current (I_L) waveforms on the oscilloscope. The tuning was performed at a DC bus voltage of $V_{DC} = 130\,V$ and under a hysteresis controlled current threshold between $-0.05\,A$ and $0.25\,A$ during charging, and between $-0.25\,A$ and $0.05\,A$ during discharging in 2L mode. In 2L mode, only switches H_2, H_6, L_0, and L_1 are involved, while all other switches remain off (Table 4.2).

Table 5.3: Optimal ZVS Transition: L_0 OFF → H_6 ON

Time	Switch	V_{SW}	V_{DS,H_6}	V_{DS,L_0}	C_{oss,H_6}	C_{oss,L_0}
$t < t_1$	L_0 ON	$\approx 0\,V$	$\approx V_{DC}$	$\approx 0\,V$	Charged	Discharged
$t_1 tot_2$	Both OFF	↑ to V_{DC}	↓ to 0 V	↑ to V_{DC}	Discharging	Charging
$t > t_2$	H_6 ON	$\approx V_{DC}$	$\approx 0\,V$	$\approx V_{DC}$	Discharged	Charged

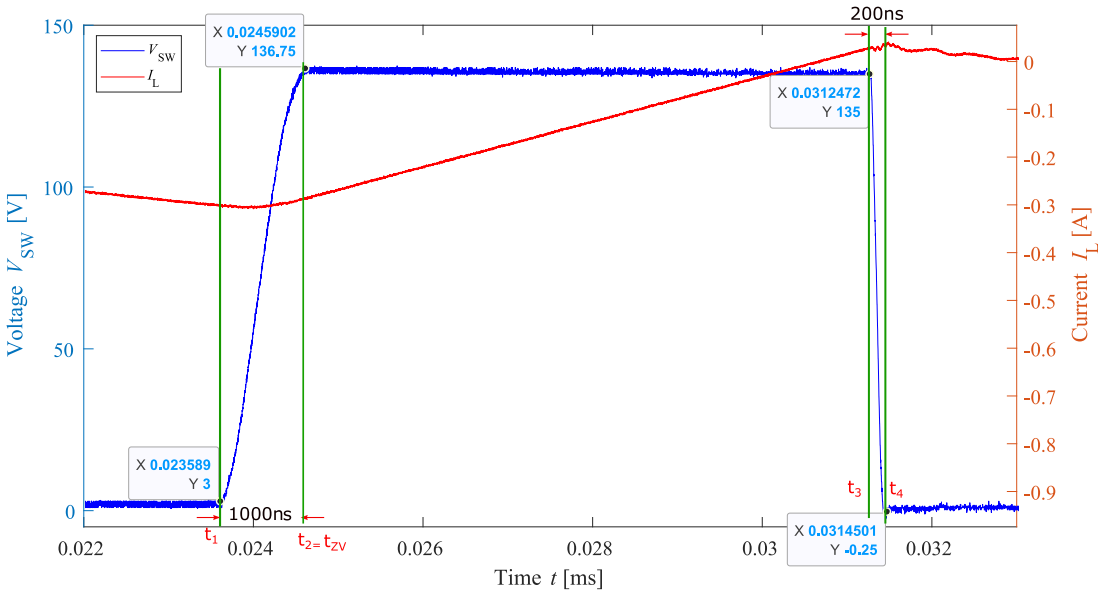


Figure 5.4: Optimised deadtime

Table 5.3 shows the optimal ZVS transition when the switch L_0 turns OFF and H_6 turns ON (V_{SW} rising transition). In the optimised ZVS (rising edge) scenario, the dead time is adjusted to match the time required ($t = t_{ZV}$) for the output capacitance (C_{oss}) of switch H_6 to fully discharge, such that the voltage across the switch reaches zero before it turns ON. This tuning results in a smooth transition of V_{SW} from approximately zero up to V_{DC} during the dead time interval $[t_1, t_2]$, enabling the subsequent H_6 switch to turn ON under zero-voltage conditions. The optimum dead time required for the rising edge is greater than that for the falling edge due to the use of higher turn ON gate resistance (R_{on}) compared to the turn OFF gate resistance (R_{off}) in both gate drivers in the hardware implementation.

For the tested operating point, the experimentally determined optimum dead times were 1000 ns for the rising edge ($t_1 - t_2$) and 200 ns for the falling edge ($t_3 - t_4$). Incorrect dead time settings can cause hard switching losses and third quadrant losses in switches.

Table 5.4: Optimal ZVS Transition: H_6 OFF → L_0 ON

Time	**Switch**	V_{SW}	V_{DS,H_6}	V_{DS,L_0}	C_{oss,H_6}	C_{oss,L_0}
$t < t_1$	H_6 ON	$\approx V_{DC}$	≈ 0 V	$\approx V_{DC}$	Discharged	Charged
$t_1 tot_2$	Both OFF	↓ to 0 V	↑ to V_{DC}	↓ to 0 V	Charging	Discharging
$t > t_2$	L_0 ON	≈ 0 V	$\approx V_{DC}$	≈ 0 V	Charged	Discharged

Hard Switching

In case of V_{SW} rising transition, the negative I_L during dead time should fully discharge the output capacitance C_{oss,H_6}, allowing H_6 to achieve ZVS. However, insufficient dead time can cause H_6 to turn on prematurely (at 600 ns instead of the ideal 1000 ns) while its drain-source voltage (V_{DS}) still high, resulting in hard switching. Since V_{DS} is non-zero and I_D is present during

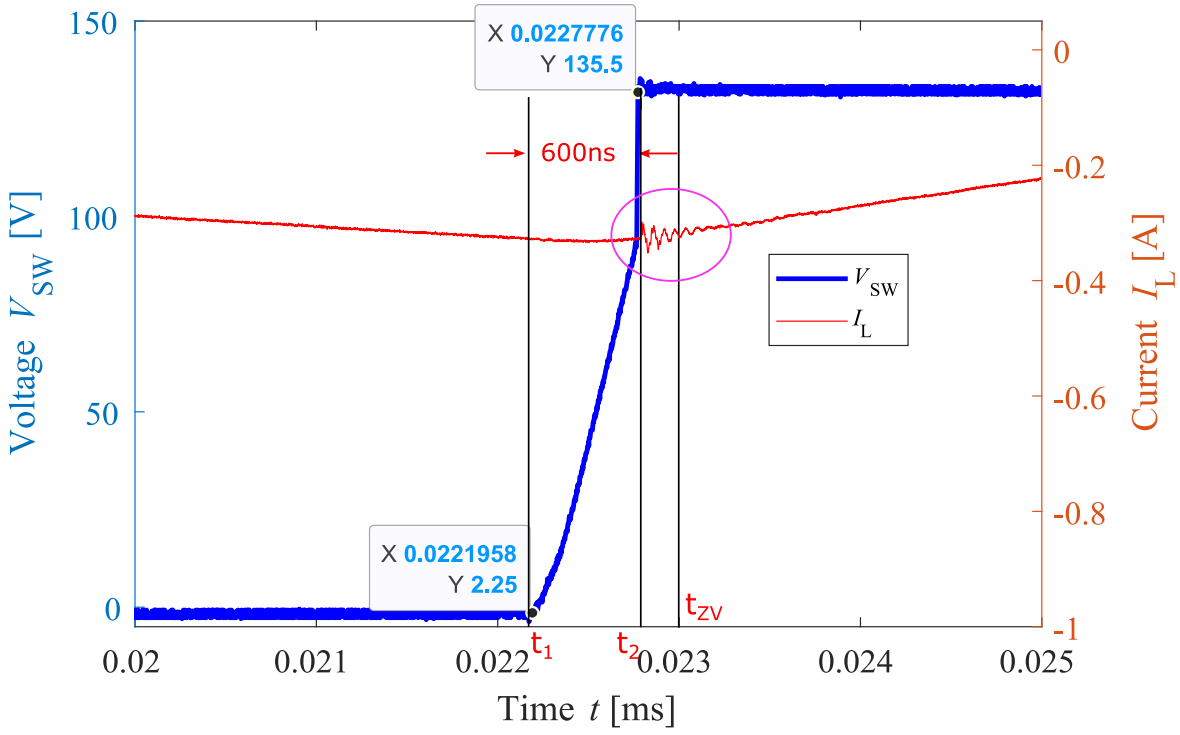


Figure 5.5: Hard switching due to insufficient dead time

the turn ON transition, a triangular overlap between voltage and current occurs, contributing to switching power loss.

This early turn ON prevents complete discharge of $C_{\text{oss,H}_6}$. As a result, the remaining capacitor energy is rapidly commutated, causing a steep voltage drop across H_6 with high dV/dt. This introduces ringing on the switch-node voltage V_{SW}. Simultaneously, the abrupt change in conduction path triggers resonant oscillations between parasitic inductances and capacitances, leading to high-frequency ringing in I_L. Figure 5.5 shows V_{SW} exhibiting a steep and noisy transition, and the I_L showing noticeable ringing when the dead time is insufficient.

Third Quadrant Loss

The negative I_L indicates that current flows from the load back to the half-bridge node during the dead time interval. With both switches OFF between t_1 and t_2, and $C_{\text{oss,H}_6}$ fully discharged around 1000 ns, the current subsequently freewheels through the body diode of H_6, which operates in the third quadrant ($V_{DS} < 0$, $I_D < 0$) for the remainder of the dead time (until 1600 ns) in case of V_{SW} rising transition.

This uncontrolled conduction path introduces a forward voltage drop and results in non-negligible power loss. The voltage overshoot observed in V_{SW} during the $L_0 \rightarrow H_6$ transition is primarily due to the reverse recovery of H_6's body diode, which injects a high di/dt current spike into the power loop. This spike induces a transient voltage across parasitic inductances, leading to the observed voltage bulge (Figure 5.6). Additional overshoot may result from resonant interactions between the inductor and switch-node capacitances.

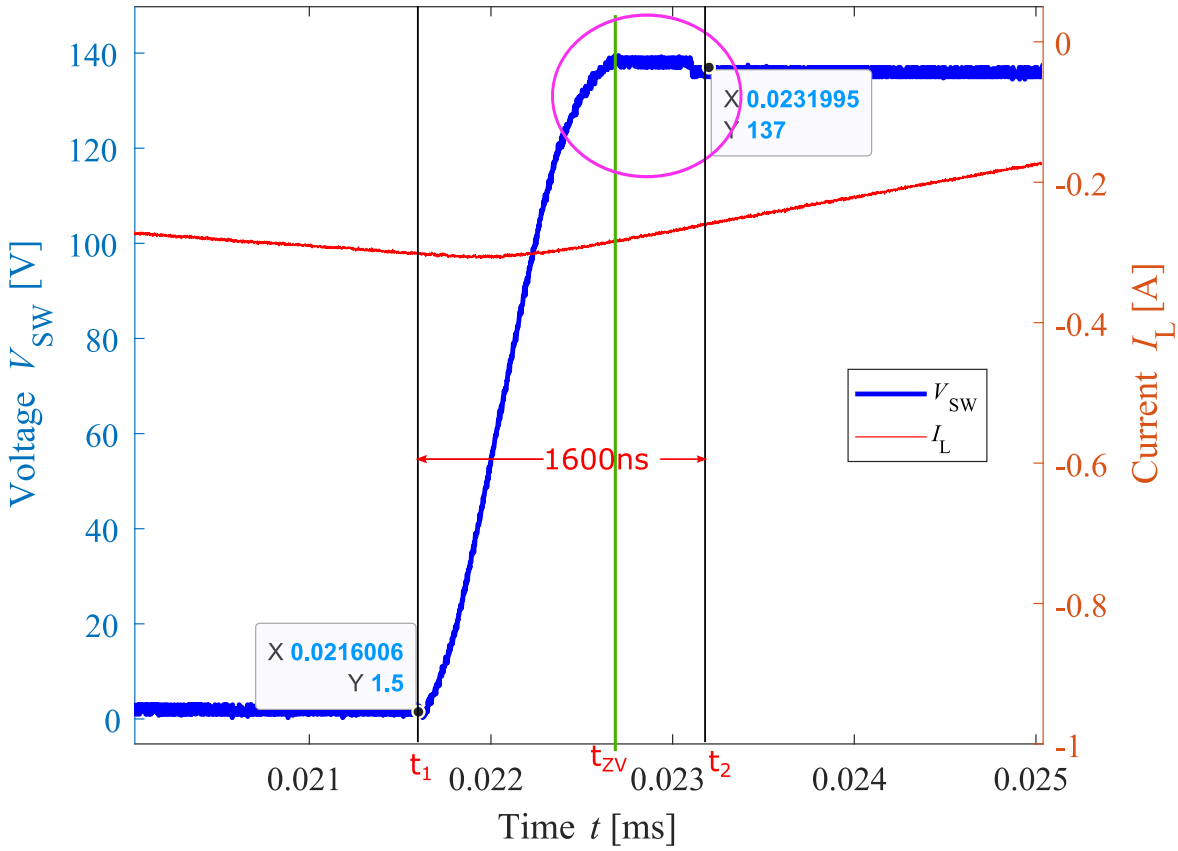


Figure 5.6: Third quadrant loss overshoot due to high dead time

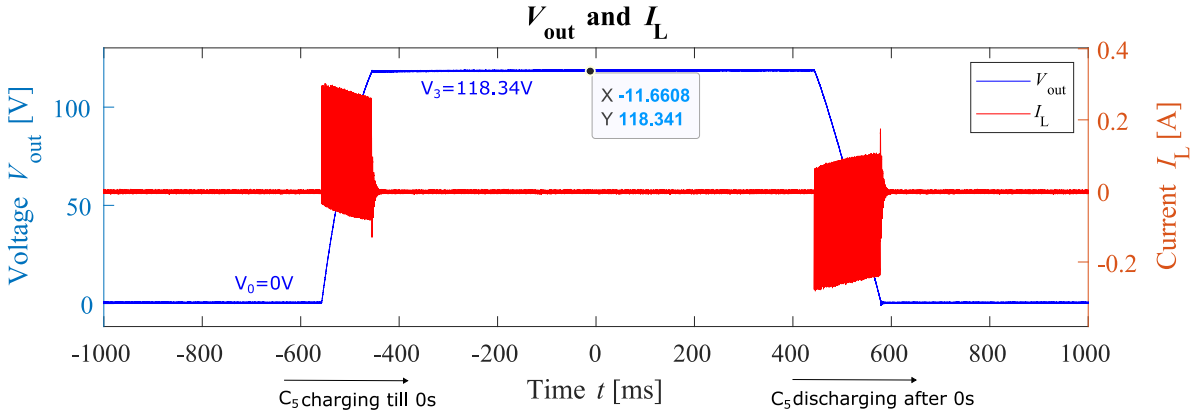


Figure 5.7: 2L Output waveforms

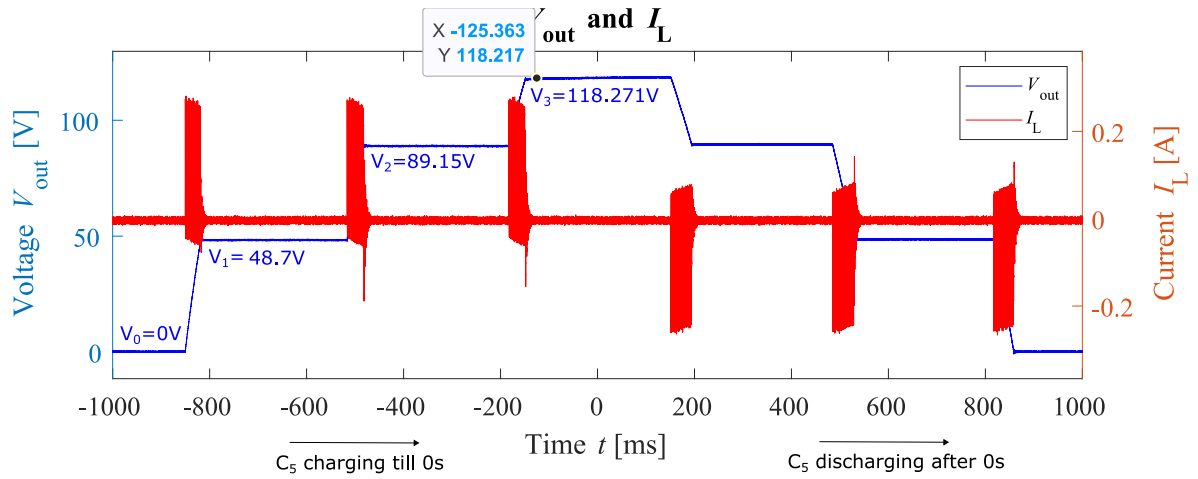


Figure 5.8: 4L Output waveforms

Table 5.5: Operating Parameters and Deadtime Comparison of 2L and 4L Converters

Parameter	**Value**
DC Input Voltage (V_{DC})	120 V
System frequency (f)	0.5 Hz
Current Hysteresis Band (Charging)	−0.05 A to 0.25 A
Current Hysteresis Band (Discharging)	−0.25 A to 0.05 A
Optimised deadtime T_1 ($t_2 - t_1$)	1000 ns (2L & 4L)
Optimised deadtime T_2 ($t_4 - t_3$)	200 ns (2L), 120 ns (4L)

5.2.3 2L and 4L Output waveforms

The 4L ANPC converter developed is operated in both 2L and four-level 4L modes at the operating point specified in Table 5.5. The dead times are optimized separately for 2L and 4L modes under a 120 V DC input voltage.

Figure 5.7 displays the output waveforms in 2L mode. The output voltage alternates between 0 V and approximately ≈ 118 V, exhibiting the characteristic 2L switching pattern. The inductor current oscillates within the predefined hysteresis bounds, indicating proper implementation of current control.

Figure 5.8 presents the output waveforms in four-level (4L) mode. The output voltage clearly shows four discrete voltage levels: 0 V, ≈ 49 V, ≈ 89 V, and ≈ 118 V. These voltage steps confirm the successful realization of multilevel operation.

Both the 2L and 4L output waveforms do not exhibit any sharp transitions or noticeable ringing, validating the use of optimally tuned dead times for minimizing switching noise and ensuring smooth transitions.

5.2.4 Efficiency Calculation

The converter input power, P_{conv}, can be estimated based on the energy required to charge the output capacitor C_5. The associated power loss is measured using a power analyzer, as shown

in Fig. 5.3. This converter power loss is measured to be

$$P_{\text{loss}} = \begin{cases} 7.9\,\text{mW}, & \text{2L} \\ 6.1\,\text{mW}, & \text{4L} \end{cases} \tag{5.1}$$

The power required to charge the output capacitor is given by:

$$P_{\text{conv}} = \frac{1}{2} \times C_{\text{out}} \times V_{\text{cap}}^2 \times 2f \tag{5.2}$$

where V_{cap} is the maximum voltage till the capacitor is charged, which is V_3 as shown in Figures. 5.7 and 5.8, and f represents the full cycle frequency.

In this case, the output capacitor C_5 has a value of $C_{\text{out}} = 50\,\mu\text{F}$, and the switching frequency is $f = 0.5\,\text{Hz}$. The factor of $2f$ is used because the capacitor charges only during half of each full cycle.

Converter Power for 2L Case

From the measurement shown in Figure 5.7, the peak capacitor voltage is $V_{\text{cap}} = 118.34$ V.

$$\begin{aligned} P_{\text{conv,2L}} &= \frac{1}{2} \times 50 \times 10^{-6} \times (118.34)^2 \times 2 \times 0.5 \\ &= 0.3501\ \text{W} \end{aligned} \tag{5.3}$$

Converter Power for 4L Case

From the measurement shown in Figure 5.8, $V_{\text{cap}} = 118.22$ V.

$$\begin{aligned} P_{\text{conv,4L}} &= \frac{1}{2} \times 50 \times 10^{-6} \times (118.22)^2 \times 2 \times 0.5 \\ &= 0.3494\ \text{W} \end{aligned} \tag{5.4}$$

Probe Losses – 2L

The probe losses for each probe connected in the 2L case are calculated and listed in Table 5.6. The RMS voltage values are obtained directly from the oscilloscope readings for each measurement channel, based on the voltage sensed by the respective probes.

$$P_{\text{loss, probe}} = \frac{V_{\text{RMS}}^2}{R_{\text{in}}} \tag{5.5}$$

Table 5.6: Measurement Probe Characteristics and Power Losses (2L Mode)

Probe	**Input Resistance (Ω)**	**Voltage RMS (V)**	**Power Loss (W)**
Keysight 10076C (V_{sw})	6.67×10^{7}	86.43	1.12×10^{-4}
TPP0850 (V_{out})	4.00×10^{7}	83.46	1.74×10^{-4}
TPP1000 (V_1)	1.00×10^{7}	12.23	1.50×10^{-5}
TPP1000 (V_2)	1.00×10^{7}	17.99	3.24×10^{-5}
Power Analyzer (V_{DC})	2.20×10^{7}	120.00	6.55×10^{-4}

For instance,

$$P_{\text{loss, TPP1000 , 2L}} = \frac{(12.23)^2}{1.00 \times 10^7} = \frac{149.5129}{1.00 \times 10^7} = 1.495 \times 10^{-5}\ \text{W} \tag{5.6}$$

So, the total probe losses for 2L can be given by,

$$\begin{aligned} P_{\text{probe, 2L}} &= 1.12 \times 10^{-4} + 1.74 \times 10^{-4} + 1.50 \times 10^{-5} + 3.24 \times 10^{-5} + 6.55 \times 10^{-4} \\ &= 0.0009884\ \text{W} \\ &= 0.99\ \text{mW} \end{aligned} \tag{5.7}$$

Probe Losses – 4L

The probe losses for each probe connected in the 4L case are calculated in a similar way and listed in Table 5.7.

Table 5.7: Measurement Probe Characteristics and Power Losses (4L Mode)

Probe	**Input Resistance (Ω)**	**Voltage RMS (V)**	**Power Loss (W)**
Keysight 10076C (V_{sw})	6.67×10^{7}	76.21	8.71×10^{-5}
TPP0850 (V_{out})	4.00×10^{7}	76.03	1.45×10^{-4}
TPP1000 (V_1)	1.00×10^{7}	39.18	1.54×10^{-4}
TPP1000 (V_2)	1.00×10^{7}	93.75	8.79×10^{-4}
Power Analyzer (V_{DC})	2.20×10^{7}	132.60	7.99×10^{-4}

$$\begin{aligned} P_{\text{probe, 4L}} &= 8.71 \times 10^{-5} + 1.45 \times 10^{-4} + 1.54 \times 10^{-4} + 8.79 \times 10^{-4} + 7.99 \times 10^{-4} \\ &= 0.0020641\ \text{W} \\ &= 2.06\ \text{mW} \end{aligned} \tag{5.8}$$

Power Loss Without Probe Losses – 2L

$$P_{\text{loss,wo probe}} = P_{\text{loss}} - P_{\text{probe}} \tag{5.9}$$

$$= 7.9 \times 10^{-3} - 9.884 \times 10^{-4} \tag{5.10}$$

$$= 6.9116 \times 10^{-3}\ \text{W} = 6.9116\ \text{mW} \tag{5.11}$$

Power Loss Without Probe Losses – 4L

$$P_{\text{loss,wo probe}} = P_{\text{loss}} - P_{\text{probe}} \tag{5.12}$$
$$= 6.1 \times 10^{-3} - 2.0641 \times 10^{-3} \tag{5.13}$$
$$= 4.0359 \times 10^{-3}\ \text{W} = 4.0359\ \text{mW} \tag{5.14}$$

2L Efficiency

$$\eta_{2L} = \left(1 - \frac{P_{\text{loss,wo probe}}}{P_{\text{conv}}}\right) \times 100\% \tag{5.15}$$
$$= \left(1 - \frac{6.9116 \times 10^{-3}}{0.3501}\right) \times 100\% \tag{5.16}$$
$$= 98.03\% \tag{5.17}$$

4L Efficiency

$$\eta_{4L} = \left(1 - \frac{P_{\text{loss,wo probe}}}{P_{\text{conv}}}\right) \times 100\% \tag{5.18}$$
$$= \left(1 - \frac{4.0359 \times 10^{-3}}{0.3494}\right) \times 100\% \tag{5.19}$$
$$= 98.84\% \tag{5.20}$$

After excluding probe losses, the effective power loss was calculated to be 6.912 mW in the 2L mode and 4.036 mW in the 4L mode, corresponding to a relative power loss reduction of approximately 41.63% in the 4L configuration.

5.2.5 Buffer capacitor selection

In a 4L ANPC converter, the DC-link buffer capacitors C_{buffer} generate intermediate voltage levels. It is important to select buffer capacitors such that $C_{\text{buffer}} \gg C_{\text{OUT}}$ to ensure negligible voltage fluctuations on the buffer capacitors during switching. When $C_{\text{buffer}} \leq C_{\text{OUT}}$, significant voltage deviations occur during voltage level transitions. The voltage change across the buffer capacitors during switching transitions is explained in Table 5.8.

If the buffer capacitance is equal or smaller than the load capacitance, a significant amount of energy is drawn from the buffer capacitors to charge the load capacitors. Moreover, it cannot store sufficient energy to support rapid current transients, leading to significant voltage drops. This occurs due to charge imbalance and limited energy storage, where the voltage across the capacitor drops rapidly according to $\Delta V = \frac{I\Delta t}{C}$ [18]. As a result, the system experiences increased ripple, degraded voltage regulation, and potential step distortions in the output waveform.

Figure 5.9 and Figure 5.10 illustrate the voltage waveforms across the buffer capacitors (C_1, C_2, and C_3, each rated at 110 μF) for two output capacitance values: $C_{\text{OUT}} = 50\,\mu\text{F}$ and $C_{\text{OUT}} =$

Table 5.8: Voltage Behavior of DC-Link Capacitors in 4-Level ANPC Converter

Capacitor	**Voltage Behavior Description**
C_1 (Bottom)	• Maintains maximum voltage when $V_{sw} = 0$, as it is not part of the conduction path. • Discharges stepwise during the $V_0 \rightarrow V_1$ and $V_2 \rightarrow V_3$ transitions. • Supplies load current during these transitions, causing gradual voltage drop. • Reaches minimum voltage when $V_{sw} = V_{dc}$, due to cumulative discharge.
C_2 (Middle)	• Voltage increases during $V_0 \rightarrow V_1$, likely due to redistribution or reverse current charging. • Exhibits minimal voltage change from $V_1 \rightarrow V_2$, indicating balanced conduction between both neutral points. • Discharges during the $V_2 \rightarrow V_3$ transition due to power devlivery to load.
C_3 (Top)	• Holds maximum voltage when $V_{sw} = V_{dc}$, as it is fully charged. • Discharges as V_{sw} decreases from $V_3 \rightarrow V_2 \rightarrow V_1$. • Recovers voltage when not in conduction path at lower levels (e.g., V_0).

$5\,\mu$F. It can be observed that for $C_{OUT} = 50\,\mu$F, the voltage fluctuations (ripples) across the buffer capacitors are more significant compared to the case with $C_{OUT} = 5\,\mu$F, where the output capacitance is ten times smaller than the buffer capacitance. These fluctuations may induce step distortions in the output voltage, increase total harmonic distortion (THD), and potentially impose overvoltage stress on the switching devices.

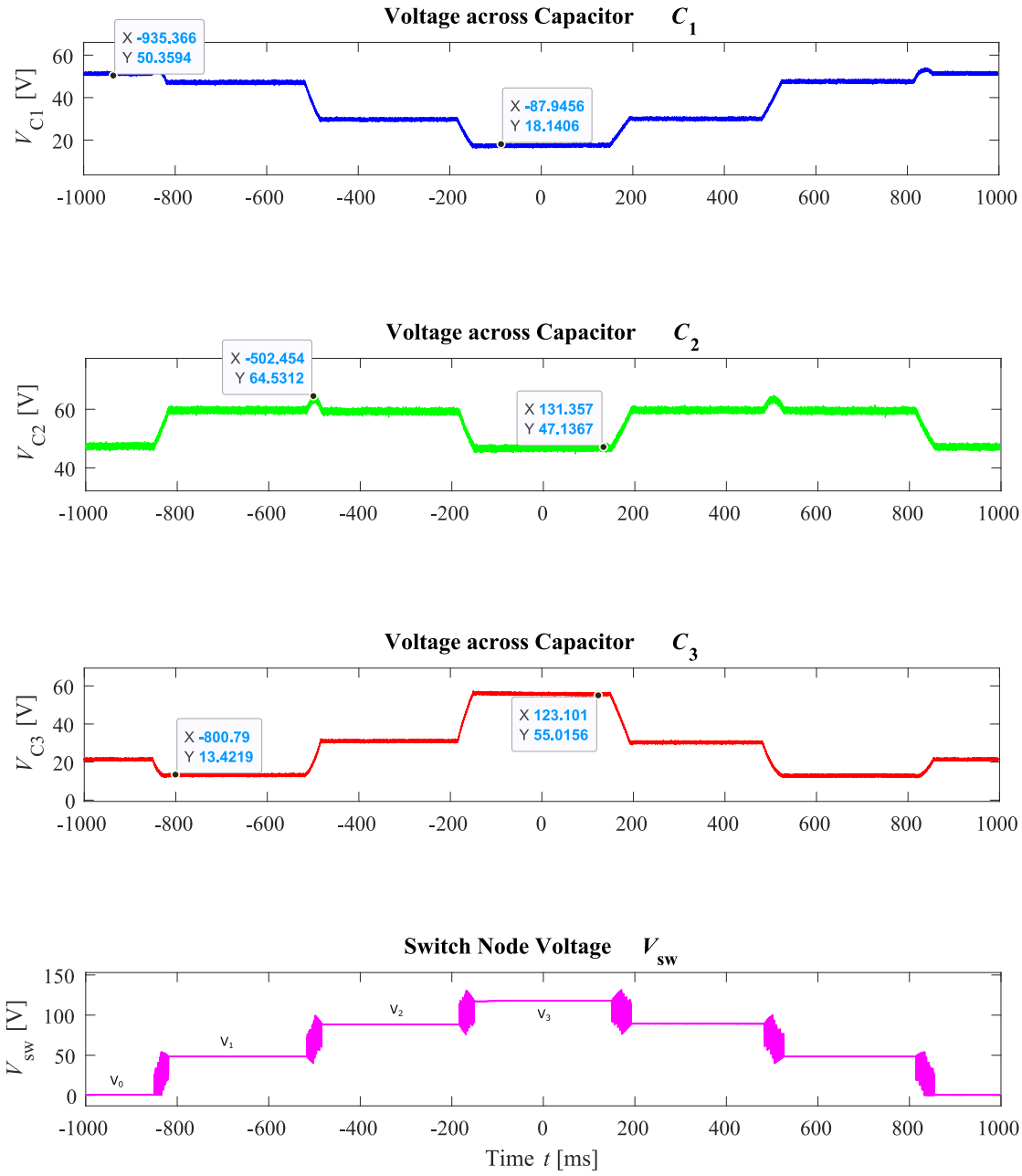


Figure 5.9: Voltage across buffer capacitors with $C_{\text{OUT}} = 50\ \mu\text{F}$

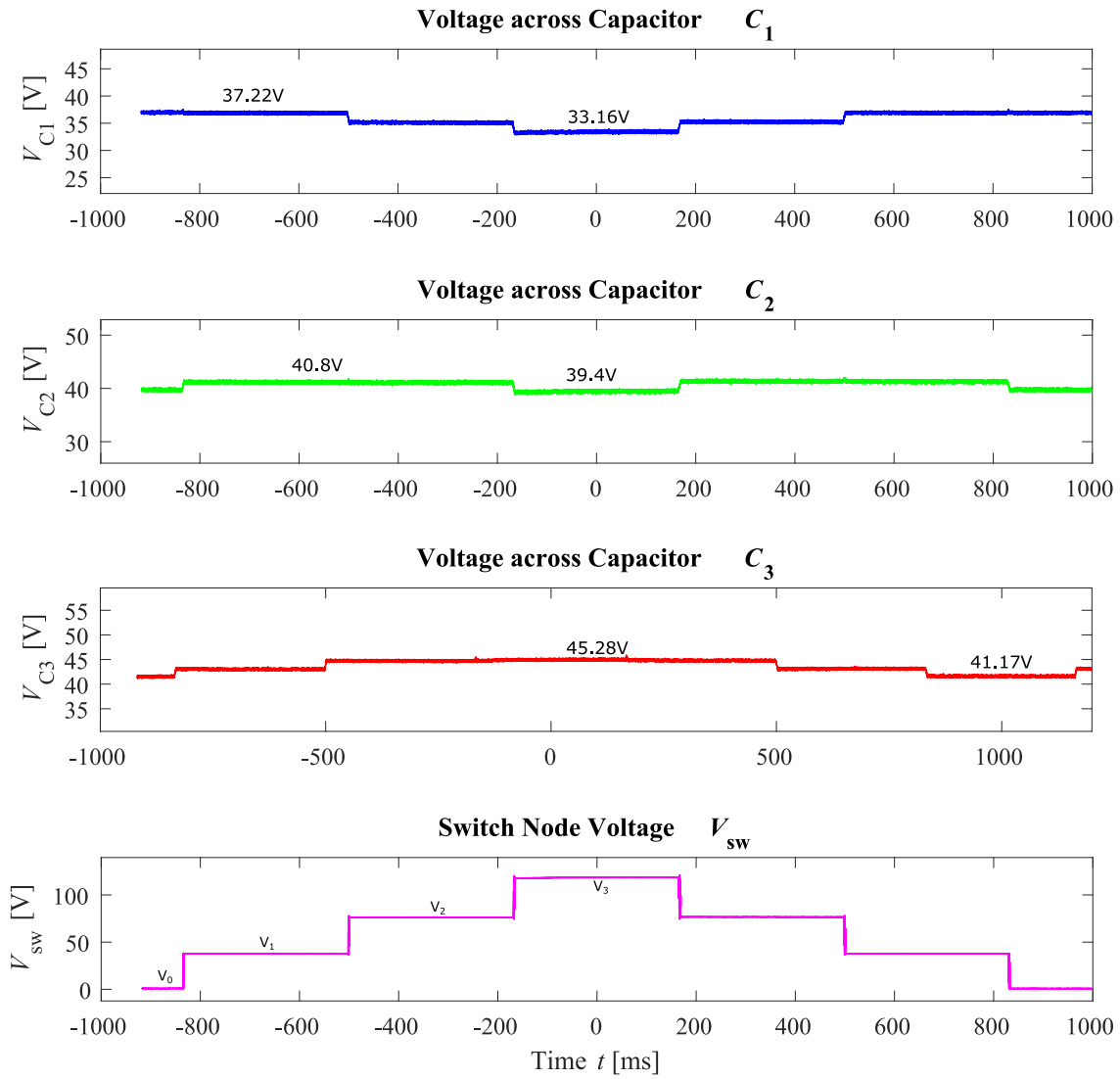


Figure 5.10: Voltage across buffer capacitors with $C_{\text{OUT}} = 5\ \mu\text{F}$

6 Summary and Conclusion

6.1 Summary

This project successfully demonstrates the design, simulation, and experimental validation of a 1.2 kV hybrid GaN/SiC 4L ANPC converter, aimed at addressing key challenges in high voltage power conversion. Each of the three major research questions concerning topology selection, complexity in control and PCB design of the 4L ANPC converter, was addressed with successful outcomes. Through careful analysis of different NPC topologies, the 4L ANPC was selected for its optimal balance of performance and component count, and for enabling hybrid GaN/SiC integration with reduced voltage blocking requirements across switches compared to DC-NPC and π-type NPC structures. Four GaN switches (GS-065-011-1-L) are utilized in low-voltage switching positions to make use of their fast switching capabilities and lower power loss, while two SiC switches (IMBG120R116M2H) are employed in high-voltage blocking positions. A custom hysteresis two-level current control algorithm was implemented, simulated, and deployed on an integrated SoC, enabling real-time dynamic frequency switching.

A compact two-layer PCB was developed for the 4L ANPC converter using innovative vertical power loop layouts to minimize parasitic inductance, along with other robust PCB techniques like voltage isolation between tracks, minimal interference between power and signal lines, optimized component placement strategies, etc. Precise dead-time optimization, appropriate gate resistance selection, and ZVS were applied to further reduce switching losses and mitigate transients. The optimum dead times were determined to be 1000 ns (V_{sw} rise) and 200 ns (V_{sw} fall) for the 2L operation, and 1000 ns (V_{sw} rise) and 120 ns (V_{sw} fall) respectively for the 4L operation at 120V DC input voltage, through manual dead-time adjustment based on the examined output voltage and output current waveforms.

Experimental tests validated the simulation results by demonstrating four distinct voltage levels in the output voltage during 4L operation, along with ZVS achieved through optimized dead-time settings. It is found that the 4L operation is achieving a higher efficiency of 98.84% compared to 98.03% in the 2L operation. After excluding probe losses, the effective power loss was calculated to be 6.912 mW in the 2L mode and 4.036 mW in the 4L mode, corresponding to a relative power loss reduction of approximately 41.63% in the 4L configuration. This successfully verifies the superiority of 4L compared to 2L operation.

6.2 Conclusion

While the system achieved satisfactory efficiency, a notable voltage ripple is observed in the buffer capacitor voltages (C_1, C_2, and C_3). This ripple arises due to the use of buffer capacitors with capacitance values closer to the output capacitance (50 µF). This effect is confirmed by

comparing the results with an experiment using a lower output capacitance of 5 μF, where the voltage ripple is significantly reduced.

Since the experiments were conducted with a near-ideal capacitive load, buffer capacitor voltage imbalance issues were not encountered, and the internal voltage levels V_1 and V_2 remained stable over time. However, in practical applications involving lossy capacitive loads (e.g., nonlinear electrocaloric pump components), the internal voltages V_1 and V_2 may decay over time due to uneven charge distribution across the capacitors [29]. This degradation can lead to a collapse of intermediate voltage levels, effectively reducing the converter's operation from four levels to two levels. To address this issue, the gate signal logic may need to be adapted to account for voltage balancing, or alternatively, advanced modulation strategies such as *Simplified Virtual Vector PWM (SVVPWM)* could be incorporated [30]. Such enhancements would promote voltage stability, support high power factor operation, improve switch protection, and help maintain high waveform quality through reduced distortion.

The proposed system architecture can be scaled for operation at even higher voltages, such as 1.7 kV, by simply replacing the 1.2 kV-rated SiC switches (used in high-voltage blocking positions) with 1.7 kV-rated SiC devices, while retaining the existing 650 V GaN switches in the lower voltage stressed positions, proving the scalabilty of the designed converter. Overall, this work lays a strong foundation for the development of next-generation multilevel converters for electric vehicles, renewable energy systems, and high efficiency electrocaloric heat pump applications.

Bibliography

[1] J. Rodriguez, J.-S. Lai, and F. Z. Peng, "Multilevel inverters: A survey of topologies, controls, and applications," *IEEE Transactions on Industrial Electronics*, vol. 49, no. 4, pp. 724–738, 2002.

[2] S. K. et al., "Recent advances and industrial applications of multilevel converters," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 8, pp. 2553–2580, 2010.

[3] M. Malinowski, K. Gopakumar, J. Rodriguez, and M. A. Perez, "A survey on cascaded multilevel inverters," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 7, pp. 2197–2206, 2010. DOI: 10.1109/TIE.2009.2030767.

[4] L. G. F. et al., "The age of multilevel converters arrives," *IEEE Industrial Electronics Magazine*, vol. 6, no. 2, pp. 28–39, 2012.

[5] H. Abu-Rub, J. Wang, Y. Huang, and F. Z. Peng, "Medium-voltage multilevel converters—state of the art, challenges, and requirements in industrial applications," *IEEE Transactions on Industrial Electronics*, vol. 57, no. 8, pp. 2581–2596, 2010. DOI: 10.1109/TIE.2009.2037087.

[6] J. M. et al., "Electrocaloric heat pumps: State of the art and future perspectives," *Applied Physics Reviews*, vol. 1, no. 1, p. 011 103, 2014.

[7] A. Krishna R and L. P. Suresh, "A brief review on multi level inverter topologies," in *2016 International Conference on Circuit, Power and Computing Technologies (ICCPCT)*, 2016, pp. 1–6. DOI: 10.1109/ICCPCT.2016.7530373.

[8] A. Nabae, I. Takahashi, and H. Akagi, "A new neutral-point-clamped pwm inverter," *IEEE Transactions on Industry Applications*, vol. IA-17, no. 5, pp. 518–523, 1981.

[9] X. Yuan, "A four-level π-type converter for low-voltage applications," in *2015 17th European Conference on Power Electronics and Applications (EPE'15 ECCE-Europe)*, 2015, pp. 1–10. DOI: 10.1109/EPE.2015.7309210.

[10] S. Jung, H.-G. Lee, and S.-K. Choi, "Comparative evaluation of three-level anpc inverter with si and sic devices for photovoltaic applications," *IEEE Transactions on Power Electronics*, vol. 31, no. 5, pp. 3796–3809, 2016.

[11] S. Mönch *et al.*, "From two to seven level gan π-type converter operation for increased efficiency of capacitive-load charging," in *2024 IEEE 11th Workshop on Wide Bandgap Power Devices & Applications (WiPDA)*, 2024, pp. 1–5. DOI: 10.1109/WiPDA62103.2024.10773264.

[12] J. Wang, I. Laird, X. Yuan, and W. Zhou, "A 1.2 kv 100 kw four-level anpc inverter with sic power modules and capacitor voltage balance for ev traction applications," in *2021 IEEE Energy Conversion Congress and Exposition (ECCE)*, 2021, pp. 5202–5209. DOI: 10.1109/ECCE47101.2021.9595038.

[13] J. Azurza Anderson, G. Zulauf, P. Papamanolis, S. Hobi, S. Mirić, and J. W. Kolar, "Three levels are not enough: Scaling laws for multilevel converters in ac/dc applications," *IEEE Transactions on Power Electronics*, vol. 36, no. 4, pp. 3967–3986, 2021. DOI: 10.1109/TPEL.2020.3018857.

[14] G. Ortiz, O. Deblecker, and J. W. Kolar, "Evaluation of the active-neutral-point-clamped (anpc) topology in modular multilevel converters," *IEEE Transactions on Industrial Electronics*, vol. 63, no. 11, pp. 6782–6793, 2016.

[15] G. Systems, *Gs-065-011-1-l 650v e-mode gan transistor datasheet (rev. 200423)*, https://gansystems.com/wp-content/uploads/2020/04/GS-065-011-1-L-DS-Rev-200423.pdf, Accessed: 2025-06-08, 2020.

[16] I. T. AG, *Imbg120r116m2h datasheet*, https://www.infineon.com/dgdl/Infineon-IMBG120R116M2H-DataSheet-v01_30-EN.pdf?fileId=8ac78c8c8d1b852e018d22003d3e20d9, Revision v01.30, Accessed: June 2025, 2023.

[17] I. T. AG, *Imbf170r450m1 datasheet*, https://www.infineon.com/dgdl/Infineon-IMBF170R450M1-DataSheet-v01_00-EN.pdf?fileId=5546d462712ef9b70171820c850e1ad9, Revision v01.00, Accessed: June 2025, 2021.

[18] T. Instruments, "Input and output capacitor selection," Texas Instruments, Application Note SLTA055, 2006, Accessed: June 2025. [Online]. Available: https://www.ti.com/lit/pdf/slta055.

[19] M. Integrated, *Max22420, dual-channel digital isolator*, Datasheet, 2019. [Online]. Available: https://www.maximintegrated.com/en/products/interface/isolation/MAX22420.html.

[20] ON Semiconductor, *Fan3111esx, high speed single-channel mosfet driver*, Datasheet, 2016. [Online]. Available: https://www.onsemi.com/pdf/datasheet/fan3111.pdf.

[21] O. Hach, "How to select gate resistors," in *4th Power Conversion North Sea (PCNS) Symposium*, Paper presented by Vishay, PCNS 2023, Sonderborg, Denmark, Sep. 2023. [Online]. Available: https://i.eu/wp-content/uploads/2023/11/5_1_PCNS-4th-Symposium-SDU-September-2023-Vishay-How-to-select-Gate-Resistors.pdf.

[22] E. P. Conversion, "Fet drivers and layout considerations," Efficient Power Conversion (EPC), White Paper WP008, 2016, Describes gate resistor and antiparallel diode techniques for EMI and dv/dt control.

[23] A. Inc., *Mcr1101-5-5 fully integrated hall effect current sensor datasheet, rev. e*, https://www.mouser.de/datasheet/2/940/6020-1101-01_RevE_MCR1101_20190627-1622836.pdf, Accessed: 2025-06-08, 2019.

[24] B. Elektronik, *Micsig cp1003b – ac/dc current probe*, https://www.batronix.com/versand/messtechnik/tastkoepfe/Micsig-CP1003B.html, Accessed: June 12, 2025.

[25] K. Armstrong, "Advanced pcb design and layout for emc - part 4: Reference planes for 0v and power," *EMC & Compliance Journal*, pp. 34–43, Jul. 2004, Available as PDF via Cherry Clough Consultants.

[26] D. Reusch and J. Strydom, "Understanding the effect of pcb layout on circuit performance in a high frequency gallium nitride based point of load converter," in *2013 Twenty-Eighth Annual IEEE Applied Power Electronics Conference and Exposition (APEC)*, 2013, pp. 649–655. DOI: 10.1109/APEC.2013.6520279.

[27] B. Sun, K. L. Jørgensen, Z. Zhang, and M. A. E. Andersen, "Research of power loop layout and parasitic inductance in gan transistor implementation," *IEEE Transactions on Industry Applications*, vol. 57, no. 2, pp. 1677–1687, 2021. DOI: 10.1109/TIA.2020.3048641.

[28] S. Moench *et al.*, "A gan-based dc-dc converter with zero voltage switching and hysteretic current control for 99% efficient bidirectional charging of electrocaloric capacitive loads," in *PCIM Europe 2022; International Exhibition and Conference for Power Electronics, Intelligent Motion, Renewable Energy and Energy Management*, 2022, pp. 1–10. DOI: 10.30420/565822251.

[29] I. Bennour, "Balancing of a dc-dc multilevel converter for the electrocaloric heat pump," M.Sc. thesis, University of Stuttgart, Stuttgart, Germany, 2024.

[30] M. A. Qamar, A. Filba-Martinez, S. Busquets-Monge, and K. Wang, "Capacitor voltage balancing of four-level anpc and pi-type converters based on simplified virtual-vector pwm," in *2023 IEEE International Conference on Electrical Systems for Aircraft, Railway, Ship Propulsion and Road Vehicles and International Transportation Electrification Conference (ESARS-ITEC)*, IEEE, 2023, pp. 1–7. DOI: 10.1109/ESARS-ITEC57127.2023.10114865. [Online]. Available: https://doi.org/10.1109/ESARS-ITEC57127.2023.10114865.

Erklärung

Ich versichere,
dass ich die Arbeit selbstständig verfasst habe,
dass ich keine anderen als die angegebenen Quellen benutzt habe und ich alle wörtlich oder sinngemäß aus anderen Werken übernommenen Aussagen als solche gekennzeichnet habe,
dass die eingereichte Arbeit weder vollständig noch in wesentlichen Teilen Gegenstand eines anderen Prüfungsverfahrens gewesen ist,
dass ich die Arbeit noch nicht veröffentlicht habe und dass das elektronische Exemplar mit dem gedruckten Exemplar übereinstimmt.

Ich erkläre mich damit einverstanden, dass die bibliographischen Daten (Titel, Autor, Erscheinungsjahr, etc.) meiner Arbeit auf der Institutswebseite des Instituts für elektrische Energiewandlung (IEW) veröffentlich werden dürfen.

Der Universität Stuttgart übertrage ich ein kostenloses, zeitlich und örtlich unbeschränktes, einfaches Nutzungsrecht meiner Arbeit und den erzeugten Arbeitsergebnissen für Zwecke der Forschung und der Lehre.

Stuttgart, 13.06.2025 Karthikeyan Vijayakumar